

Mapping AI Literacy Research: A Bibliometric and Topic Modeling Study

Although AI literacy is increasingly invoked across education, healthcare, libraries, and professional training, it remains unclear whether and how this growing body of work coheres as a scholarly field. This study provides a large-scale bibliometric and topic-modeling analysis of self-identified AI literacy research, using a corpus of 2,198 Scopus-indexed documents published between 2016 and 2025. Adopting a narrow, term-based retrieval strategy, we examine dissemination patterns, authorship and country networks, and thematic structures to clarify how AI literacy is currently constituted in its own discourse. Results show that publication output is concentrated in a small number of high-productivity venues and collaborative networks, with the United States, China, Hong Kong, and the United Kingdom forming a geographic core of the field. Co-authorship networks reveal a moderately connected but increasingly diversified research community anchored by clusters in East Asia, North America, and Europe. A KMeans-guided BERTopic workflow applied to title and abstract text identifies ten thematic concentrations: educator readiness and ethical competencies; adoption factors; higher education integration; generative AI instruction; school curriculum and STEM education; ChatGPT, writing, and prompting; human decision-making and trust; children’s experiential learning; healthcare AI literacy; and libraries and information services. Temporal analysis shows a shift from earlier emphases on school curriculum, educator readiness, and children’s learning toward post-2023 growth in generative AI, ChatGPT-mediated writing, higher education integration, and adoption-oriented research. The study maps AI literacy as an emerging field and provides an empirical basis for future framework development, measurement, and context-sensitive educational design.

Introduction

Artificial intelligence (AI), “the science and engineering of making intelligent machines” (McCarthy, 2007, p. 2), has rapidly transitioned from a specialized technical innovation into a pervasive socio-technical infrastructure that now shapes knowledge production and everyday decision-making (Lintner, 2024). In little more than a decade, AI systems, once embedded primarily in industrial, scientific, or computational contexts, have become ubiquitous in search engines, social platforms, productivity tools, healthcare diagnostics, financial systems, and educational technologies (Crawford, 2021; [Pasquale, 2020](#); Sadowski, 2020). The public release of generative AI and large language models (LLMs) such as ChatGPT and Claude has accelerated this transformation dramatically. These systems have introduced new forms of automation, new epistemic uncertainties, and new cognitive affordances, prompting profound debates about the meaning of authorship, the nature of expertise, and the distribution of agency between humans and machines (Krakowski, 2025; Miao et al. 2025; Floridi & Chiriatti, 2020). As AI systems grow more powerful and more deeply integrated into daily life, scholars, policymakers, and institutions increasingly argue that people require new competencies collectively described as *AI literacy*, to

navigate, critique, understand, and influence AI-mediated environments (Ngoveni, 2025; Pinski and Benlian 2024; Biagini, 2025; Long & Magerko, 2020; Ng et al. 2021; Long et al. 2020).

Although often framed as a novel concept, AI literacy builds on long-standing traditions in digital, data, information, and media literacies (Ng et al., 2021). These traditions emphasize the skills and dispositions people need to evaluate digital information, interpret data-driven systems, engage with computation, and critique the sociotechnical forces shaping technological life. Yet AI introduces qualitatively different challenges: greater system opacity, emergent behaviors, probabilistic reasoning, embedded biases, and epistemic instability (Burrell 2016). Understanding how AI systems operate and how they shape decisions, perceptions, and opportunities now requires literacy practices that exceed earlier frameworks. As such, AI literacy has emerged quickly as both a theoretical construct and a practical imperative, invoked across K-12 education (Casal-Otero et al. 2023; Wang & Lester 2023), higher education (Michel-Villarreal et al. 2023; Zhang & Aslan 2021), healthcare (Kang et al. 2023; Kimiafar et al. 2023; Subaveerapandiyan et al. 2024), libraries (Tzanova 2024; Ali & Richardson 2025; Miltenoff, 2024), workforce development (Chetty 2023; Li & Kim 2024), and public policy (Thierer et al. 2017; Valle-Cruz et al. 2019; Nordström 2022; Tan and Tang 2025).

Despite its rapid uptake, AI literacy remains an early-stage and contested field. Existing publications define it inconsistently, applying the term to technical knowledge, ethical awareness, socio-civic dispositions, practical skills for interacting with AI tools, or combinations of these (Ng et al. 2021). Definitions vary across disciplines and regions. In K–12 contexts, AI literacy often emphasizes conceptual understanding of machine learning, classification, data flows, and algorithmic decision-making, along with ethical awareness and inquiry-based pedagogies (Casal-Otero et al. 2023). In higher education, AI literacy has expanded to include competencies related to generative models, academic integrity, disciplinary uses of AI, and professional readiness (Southworth et al., 2023; Chan, 2023). In healthcare, AI literacy is framed as a safety-critical competency linked to diagnostic reasoning, clinical judgment, and responsible use of AI-supported systems (Figaredo & Stoyanovich 2023; Stamboliev 2023; Pramod, 2025). Libraries and information science researchers approach AI literacy from the perspective of information stewardship, instructional mediation, and ethical information practices (Pu et al. 2024; LaFlamme, 2025; Monyela & Tella, 2024). Across these domains, AI literacy is conceptualized through distinctive epistemological commitments and institutional priorities (Xi et al. 2025; Ng et al. 2021).

As a result, the AI literacy literature has grown rapidly but unevenly. Existing reviews have synthesized important strands of this work in K–12 education, higher education, healthcare, nursing, and information science, but most remain bounded by sector, population, or application context (Pinski & Benlian, 2024; Casal-Otero et al., 2023). Other reviews examine AI literacy alongside adjacent constructs such as computational thinking, AI education, digital literacy, and data literacy, which makes it difficult to distinguish how AI literacy is developing as a scholarly discourse in its own right (Jantakun et al., 2025). Although recent reviews and bibliometric studies

have begun to document the field's fragmentation, they generally cover shorter timeframes, smaller or more domain-specific corpora, or periods before the widespread uptake of generative AI (Yang et al., 2025). As a result, it remains unclear how self-identified AI literacy research is organized across publication venues, authorship communities, geographic regions, collaboration networks, and thematic concentrations. This study addresses that gap by mapping Scopus-indexed AI literacy research through 2025 and combining bibliometric, collaboration, geographic, and embedding-based thematic analyses.

This study responds to that gap by presenting a large-scale bibliometric and topic-modeling analysis of research that explicitly self-identifies as AI literacy. Using a deliberately narrow, term-based retrieval strategy, the study assembles a corpus of 2,198 Scopus-indexed documents published between 2016 and 2025 that contain the terms “AI literacy,” “A.I. literacy,” or “artificial intelligence literacy.” By focusing exclusively on self-identified AI literacy research, the study treats AI literacy as a scholarly discourse in its own right, rather than inferring its presence from related constructs such as AI education, computational thinking, or digital literacy. This approach provides a clear empirical basis for examining how the field defines itself, where it is disseminated, and how its thematic organization has evolved, particularly in response to the rapid uptake of generative AI. The study maps dissemination patterns across journals, conferences, and other publication formats; examines authorship and country-level collaboration networks; and identifies thematic structures using bibliometric analysis, network analysis, and embedding-based topic modeling. For the topic-modeling component, a cleaned subset of 2,120 records with title and abstract text was analyzed using a KMeans-guided BERTopic workflow, which combined sentence-transformer embeddings, fixed-cluster assignment, and class-based TF-IDF topic representation. These methods enable analysis of AI literacy as both a social and semantic field, revealing patterns of publication, collaboration, geographic concentration, and thematic organization within the literature.

This study makes three contributions. First, it provides a field-level account of AI literacy as a self-identified research area, clarifying how scholarship that explicitly uses the AI literacy label differs from broader work on AI education, digital literacy, data literacy, and computational thinking. Second, it offers an empirical baseline for future AI literacy research by identifying the field's main venues, collaboration structures, geographic centers, and thematic concentrations across a large Scopus-indexed corpus. Third, it shows how the rise of generative AI has altered the conceptual scope of AI literacy, extending it from foundational knowledge and curriculum design toward situated practices of AI use, evaluation, trust, writing, prompting, and professional judgment.

The remainder of this paper is organized as follows. The next section reviews AI literacy research. It situates the study's field-level mapping rationale, frames the mapping approach as a bibliographic review, presents the study's analytic logic, and states the four research questions guiding the analysis. The third section describes the data sources and methods used to address each

research question. The following two sections present the results and discussion, and the final section concludes the paper.

AI Literacy Research

Research on AI literacy has expanded rapidly across disciplines and application contexts, reflecting sustained scholarly efforts to understand how individuals and institutions engage with artificial intelligence systems (Long & Magerko, 2020; Ng et al., 2021; Ng et al., 2024). Recent reviews have synthesized important parts of this literature, including conceptual frameworks, competency models, educational applications, sector-specific developments, and measurement approaches (Lintner, 2024; Ng et al., 2024). These reviews have clarified many dimensions of AI literacy and have shown that the construct is relevant across education, professional practice, and public life. However, they also reveal the limits of review-based synthesis in a rapidly expanding and increasingly dispersed literature. Existing reviews tend to focus on particular domains, populations, definitions, or pedagogical approaches, leaving less clear how AI literacy research is organized as a scholarly field across publication venues, collaboration networks, geographic regions, and thematic clusters.

Emergence of AI Literacy

As AI systems became embedded in everyday technologies and institutional practices, scholars argued that traditional forms of digital, data, and information literacy were insufficient for navigating algorithmic environments (Archambault et al., 2024; Ng et al., 2021; Polomoshnov & Lobanova, 2025). In response, AI literacy emerged as a construct intended to capture competencies specific to understanding, evaluating, and engaging with AI systems, extending rather than replacing existing literacy frameworks (Long & Magerko, 2020; Ng et al., 2021). Early conceptual work drew on traditions of computational thinking, media literacy, and data literacy, emphasizing foundational understanding of AI mechanisms such as machine learning, classification, prediction, and data processing (Long & Magerko, 2020; Ng et al., 2021; Tiernan et al., 2023). These studies framed AI literacy primarily as conceptual awareness of algorithmic systems and their operation, often clustered into constructs such as “know and understand,” “use and apply,” “evaluate and create,” and “AI ethics” (Ng et al., 2021).

As AI tools became more accessible and integrated into consumer-facing applications, conceptualizations of AI literacy expanded to foreground practice and context (Casal-Otero et al., 2023; Ng et al., 2024). Definitions increasingly incorporated practical competencies for interacting with AI systems such as chatbots, recommender systems, and generative models, as well as dispositions related to critical thinking, ethical awareness, and human–AI collaboration (Giustini & Dastyar, 2024; Long & Magerko, 2020; Ng et al., 2021; Ng et al., 2024). Influential reviews have articulated AI literacy as a multidimensional construct encompassing technical knowledge,

applied skills, ethical and legal awareness, critical evaluation of AI outputs, and sensitivity to social impacts, often connecting AI literacy to responsible AI and data justice agendas (Domínguez Figaredo & Stoyanovich, 2023; Lintner, 2024; Ng et al., 2024; Velandar et al., 2024). This work has helped establish AI literacy as a legitimate and complex area of inquiry, laying the conceptual foundation for sector-specific and measurement-focused research (Long & Magerko, 2020; Ng et al., 2024).

Despite this progress, definitions and emphases continue to vary across disciplines, contexts, and populations, producing a heterogeneous conceptual landscape. Much of the conceptual literature is framework-driven, focusing on articulating what AI literacy should include rather than empirically examining how different dimensions are distributed or combined across the published literature (Ng et al., 2021). As a result, while conceptual scholarship has clarified key dimensions of AI literacy, it provides limited insight into how those dimensions are prioritized across research communities, sectors, or world regions, or how they have shifted with the rise of generative AI (Almatrafi et al., 2024).

Limitations of Existing Reviews

Existing reviews are generally bounded by sector, population, or conceptual lens and frequently integrate AI literacy with adjacent constructs such as AI education, computational thinking, digital literacy, or data literacy (Casal-Otero et al., 2023; Ng et al., 2024). This integrative approach is often appropriate for addressing specific research or practice-oriented questions—for example, in K–12 computing education, teacher professional development, higher education, healthcare, libraries, or discipline-specific applications. However, it limits visibility into AI literacy as a self-identified scholarly discourse. Reviews can clarify definitions, competencies, and intervention designs, but they are less able to show how the field is organized across publication venues, authorship networks, geographic regions, and thematic communities.

This limitation matters because AI literacy research has developed unevenly across domains. In K–12 education, AI literacy often emphasizes conceptual understanding of machine learning, classification, data flows, and algorithmic decision-making, alongside ethical awareness and inquiry-based pedagogies (Casal-Otero et al., 2023; Wang & Lester, 2023). In higher education, it has expanded to include competencies related to generative models, academic integrity, disciplinary uses of AI, and professional readiness (Chan, 2023; Southworth et al., 2023). In healthcare, AI literacy is framed as a safety-critical competency linked to diagnostic reasoning, clinical judgment, and responsible use of AI-supported systems (Domínguez Figaredo & Stoyanovich, 2023; Pramod, 2025; Stamboliev, 2023). Libraries and information science researchers approach AI literacy from the perspective of information stewardship, instructional mediation, and ethical information practices (LaFlamme, 2025; Monyela & Tella, 2024; Pu et al., 2024). These domain-specific developments show the breadth of AI literacy research, but they also demonstrate why sectoral reviews alone cannot explain how the field coheres as a whole.

Earlier bibliographic studies have begun to address this gap by examining AI literacy research published up to the early 2020s (Durak et al., 2024; Tenório et al., 2023; Yang et al., 2025). These studies provide useful descriptive overviews of publication patterns prior to the widespread adoption of generative AI. However, they often rely primarily on traditional bibliometric indicators, cover limited article sets or timeframes, or focus on specific areas such as education. Consequently, they do not fully capture how conceptual emphases cluster within the literature, how different strands of AI literacy research relate to one another at a thematic level, or how the field has evolved through 2025. This limitation has become increasingly salient following the rapid expansion of generative AI after 2022, which accelerated publication activity across disciplines and prompted the emergence of new frameworks, scales, and intervention studies explicitly labeled as AI literacy (Lintner, 2024).

Field-Level Mapping as a Response

The limitations of existing reviews point to the need for a field-level mapping of AI literacy research. Rather than proposing another definition of AI literacy or evaluating specific educational or professional interventions, the present study examines how scholarship explicitly labeled as AI literacy is organized, disseminated, and thematically structured. This approach treats AI literacy as a self-identified scholarly discourse and focuses on how research using this label has emerged, circulated, and clustered within the academic literature.

A field-level mapping perspective shifts attention away from individual definitions or frameworks and toward the structural characteristics of the literature as a whole. AI literacy research is produced by diverse research communities, appears across a wide range of scholarly venues, and addresses multiple populations and application contexts. These characteristics make it difficult to apprehend the field's overall configuration through close reading or sector-specific review alone. A bibliographic mapping approach, by contrast, allows the contours of the field to be observed through aggregate patterns: where research is disseminated, who contributes to it, how collaboration is organized, how geographic participation is distributed, and how thematic priorities cluster and shift over time (Donthu et al., 2021; López-Pernas et al., 2023).

Central to this approach is the decision to treat AI literacy as a self-identified scholarly discourse. The analysis focuses exclusively on publications that explicitly use the term “AI literacy,” “artificial intelligence literacy,” or “A.I. literacy.” This delimitation does not imply that other forms of research on AI education, digital literacy, data literacy, or computational thinking are irrelevant. Rather, it reflects a deliberate analytical choice to examine how the field organizes itself around the label AI literacy and how that label functions as a marker of scholarly identity. By focusing on self-identification, the study avoids imposing external categorizations and instead examines the field on its own terms.

Mapping AI literacy research as a self-identified discourse makes it possible to examine several interrelated dimensions of the field. Dissemination patterns show where AI literacy research is published and which venues serve as primary channels for scholarly communication. Authorship and collaboration patterns reveal how knowledge production is organized and which research communities shape the field. Geographic patterns show how participation and influence are distributed across countries and regions. Thematic analysis identifies how AI literacy research clusters conceptually and how those clusters have shifted over time, particularly in response to generative AI.

The mapping perspective adopted in this study is therefore descriptive and analytical rather than prescriptive. It does not evaluate the effectiveness of AI literacy interventions, assess the normative adequacy of particular frameworks, or seek to resolve definitional disagreements. Instead, it makes the structure of AI literacy research visible. By doing so, it provides a basis for understanding how different strands of work relate to one another, how scholarly attention is distributed, and how the field has responded to recent technological developments.

Guided by this analytical logic, the present study addresses the following research questions:

- *RQ1. Dissemination patterns: How is AI literacy research disseminated across scholarly venues and publication formats?*
- *RQ2. Knowledge production and collaboration: Who produces AI literacy research, and how are authors connected through co-authorship and collaboration networks?*
- *RQ3. Geographic distribution: How is AI literacy research distributed across countries and regions, and what patterns of international collaboration characterize the field?*
- *RQ4. Thematic structure: What thematic concentrations structure AI literacy research as a self-identified scholarly discourse?*

Data and Methods

This section describes the data collection strategy and analytic procedures used to map and interpret the scholarly literature on AI literacy. Because AI literacy is an interdisciplinary and rapidly changing area, the study combines descriptive bibliometrics, network analysis, and embedding-based natural language processing to capture both metadata-driven patterns of dissemination and text-driven thematic structure (Donthu et al., 2021).

Data Source and Search Strategy

The corpus was collected from Scopus in November 2025 using an intentionally narrow, term-based query designed to retrieve literature that explicitly self-identifies as AI literacy. The query searched for occurrences of the phrases “AI literacy,” “A.I. literacy,” and “artificial intelligence literacy” across titles, abstracts, author keywords, index keywords, conference names, and source

titles, constrained to publications from 2010 through 2025. The exact Scopus advanced search string used for retrieval is documented in Appendix A to ensure reproducibility.

The initial retrieval returned 2,224 records covering the period 2016–2025. During preprocessing, records missing critical bibliographic information or otherwise unsuitable for analysis were removed, yielding a final bibliometric corpus of 2,198 Scopus-indexed documents. For each record, we retained standard bibliographic metadata, including authors, affiliations, title, abstract, publication year, source title, document type, citation counts, DOI, open access status, author keywords, index keywords, and full affiliation strings. These fields were used to support the dissemination, authorship, country-level, and citation analyses.

Preprocessing proceeded in two stages. First, bibliographic metadata were cleaned and standardized for analysis. This included normalizing publication years, source titles, document types, author names, and affiliation strings where necessary to reduce inconsistencies introduced by database metadata. Country names were extracted from affiliation fields and harmonized to support geographic and collaboration analyses. Second, a separate text-preprocessing procedure was applied for the topic-modeling component. Because topic modeling relied on substantive document content, only title and abstract text were used. Records with missing, unusable, or non-substantive abstracts were excluded from the topic-modeling subset, as were conference review records and records identified as errata, retractions, or other non-research publication types. This produced a cleaned topic-modeling corpus of 2,120 records.

Analytic Methods

To address the four research questions, we used a layered analytic pipeline that paired descriptive bibliometrics with network analysis and embedding-based topic discovery. Each component was selected to capture a different dimension of the field: dissemination patterns, knowledge production and collaboration, geographic distribution, and thematic structure.

For RQ1, which concerns dissemination patterns, we computed descriptive bibliometric statistics to profile the growth and distribution of AI literacy research. These analyses included annual publication counts, publication growth over time, document-type distributions, open-access proportions, and source-title productivity. Venue-level productivity was examined by counting the number of AI literacy publications per journal, conference proceeding, book series, or other outlet. Citation-based indicators, including total citations and mean citations per document, were used to distinguish publication volume from citation visibility. These analyses made it possible to identify the venues and publication formats through which AI literacy research has been disseminated and to assess how publication activity changed over time.

For RQ2, which concerns knowledge production and collaboration, we analyzed author productivity and co-authorship patterns. Author names were standardized prior to analysis to reduce obvious duplication caused by metadata variation. A co-authorship network was then

constructed with authors as nodes and co-authored publications as weighted edges. Network measures included degree centrality, weighted degree, and betweenness centrality, allowing us to identify both highly connected authors and authors who served as bridges across collaboration clusters. Community detection was used to identify major collaboration groups within the author network. We also examined author-level publication counts and citation indicators to distinguish prolific contributors from highly cited contributors.

For RQ3, which concerns geographic distribution, we extracted country information from author affiliation strings and harmonized country names for analysis. Country-level productivity was measured using total publication counts, total citations, and mean citations per document. We also distinguished single-country publications from internationally co-authored publications. A country collaboration network was constructed with countries as nodes and co-authorship ties as weighted edges. This network was used to examine international collaboration patterns, identify central countries in the field, and assess the extent to which AI literacy research is geographically concentrated or internationally distributed.

For RQ4, which concerns thematic structure, we used an embedding-based topic-modeling workflow to identify major thematic concentrations in AI literacy research. Topic modeling was conducted on the cleaned subset of 2,120 records with usable title and abstract text. Author keywords and index keywords were retained for bibliographic description but were not used as input text for the topic model. This choice ensured that the thematic model was based on substantive document content rather than metadata labels.

The topic-modeling procedure used a KMeans-guided BERTopic workflow. First, title and abstract texts were concatenated for each record and minimally cleaned to remove non-substantive boilerplate, URLs, HTML artifacts, and excess whitespace while preserving meaningful language. Second, document embeddings were generated using the sentence-transformers model all-MiniLM-L6-v2, selected for its efficiency and strong performance on short- and medium-length academic text. Third, KMeans clustering was used to assign documents to a fixed number of thematic clusters. This guided clustering approach was adopted after preliminary HDBSCAN-based BERTopic models produced either high outlier rates or overly collapsed topic structures. Fourth, BERTopic's class-based TF-IDF representation was used to extract representative terms for each cluster and support topic interpretation.

Several topic solutions were compared, including 8-, 9-, 10-, 12-, 14-, 15-, and 18-topic models. Model selection was based on interpretability, topic-size balance, redundancy among topics, and diagnostic indicators of cluster quality. Lower-topic solutions, especially the 8- and 9-topic models, over-merged substantively distinct themes into broad education-centered categories. Higher-topic solutions produced greater granularity but introduced additional redundancy among closely related education and generative AI topics. The 10-topic solution was selected as the best balance between parsimony and thematic specificity.

The final model assigned all 2,120 topic-modeling records to ten interpretable thematic concentrations. Topic labels were developed through inspection of top c-TF-IDF terms, representative documents, topic sizes, and semantic relationships among topics. The final topics were interpreted as overlapping thematic concentrations rather than sharply separated research silos. To assess robustness and interpretability, we examined topic-size distributions, embedding-space cluster diagnostics, topic-centroid similarities, top-word overlap, and stability across repeated clustering runs. Temporal topic trajectories were then computed by aggregating topic assignments by publication year and by broader periods corresponding to the pre-generative AI period, the 2023 transition year, and the 2024–2025 generative AI acceleration period.

All analyses were conducted in Python using standard libraries for data processing, visualization, network analysis, and natural language processing. Descriptive statistics and tables were generated using pandas and NumPy. Visualizations were produced using matplotlib. Network analyses used standard graph-based procedures for centrality and community detection. Topic modeling used sentence-transformers, scikit-learn, and BERTopic.

Results

RQ1: Dissemination Patterns in AI Literacy Research

Table 1 summarizes the descriptive characteristics of the dataset. The corpus consists of 2,198 Scopus-indexed documents published between 2016 and 2025, with a total of 28,534 citations and an average of 12.98 citations per document. The literature is extremely recent, with an average document age of only 0.6 years, indicating that AI literacy has expanded most rapidly in the last few years of the observation period. This growth is also reflected in the average annual growth rate of 146.36%, suggesting that AI literacy has moved quickly from a small and emerging research area into a rapidly expanding scholarly field.

The distribution of document types shows that AI literacy research is dominated by formal scholarly outputs. Journal articles represent the largest category, accounting for 1,177 documents, followed by conference papers (n = 593), book chapters (n = 199), and review articles (n = 122). Less common formats include notes, conference reviews, editorials, and books. This distribution suggests that AI literacy research is being disseminated primarily through journal and conference channels, while also beginning to appear in broader scholarly formats such as books and edited volumes. Nearly half of all documents are open access (44.36%), indicating reasonably broad accessibility. The average number of authors per document is 3.52, suggesting that AI literacy scholarship is characterized by moderate to high collaborative intensity.

Table 1. Descriptive Characteristics of AI Literacy Research

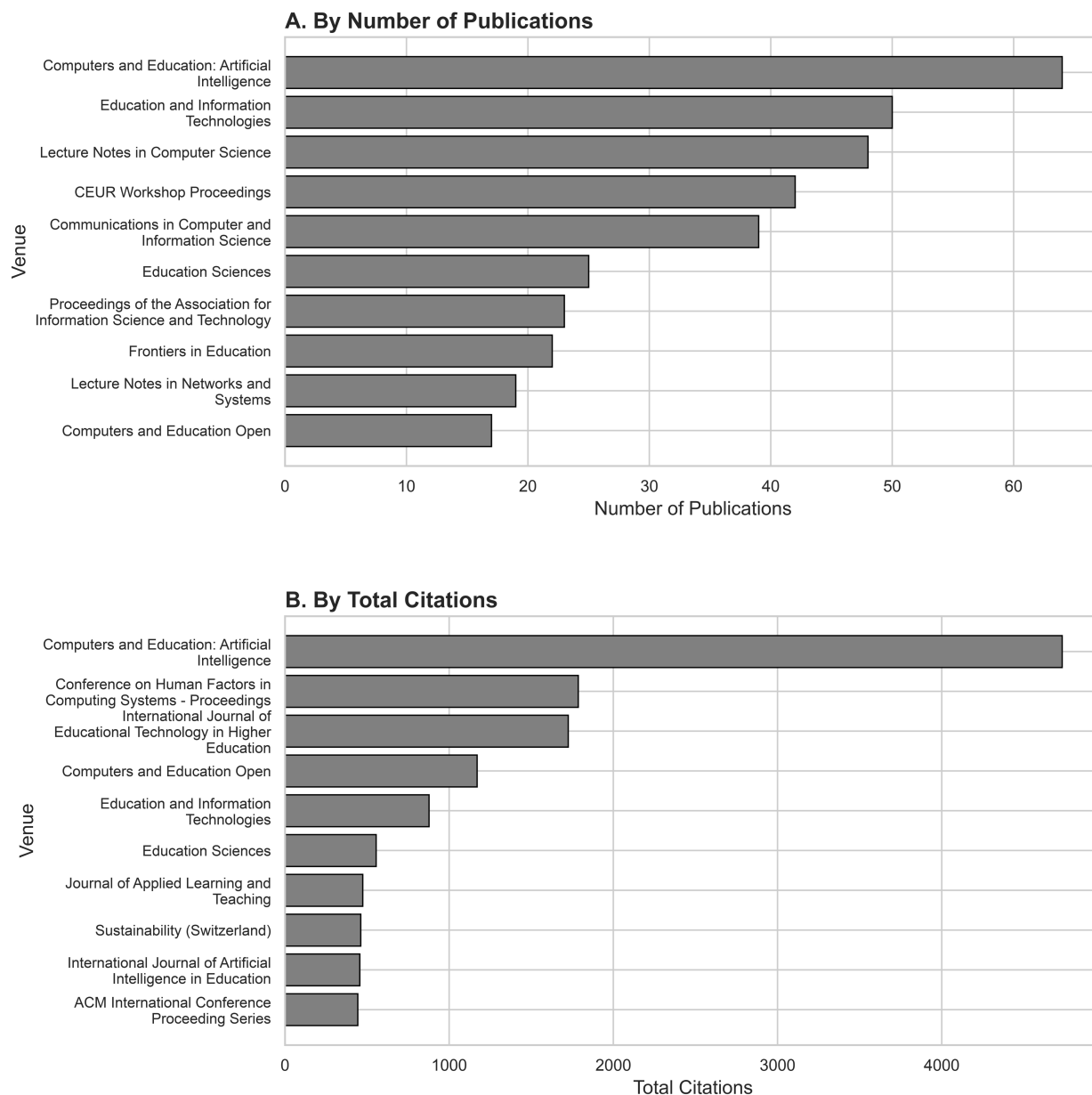
Metric	Value
Total Documents	2198

Timespan	2016–2025
Total Citations	28534
Average Citations per Document	12.98
Average Annual Growth Rate (%)	146.36
Document Types	Article: 1177 Conference paper: 593 Book chapter: 199 Review: 122; Note: 39 Conference review: 32 Editorial: 23 Book: 12
Open Access (%)	44.36
Average Authors per Document	3.52
Average Document Age (years)	0.6

The venues responsible for disseminating AI literacy research are diverse, spanning education, information science, computer science, educational technology, and human–computer interaction. Figure 1 shows the top publication venues by both number of publications and total citations. By publication volume, *Computers and Education: Artificial Intelligence* is the most productive outlet, followed by *Education and Information Technologies*, *Lecture Notes in Computer Science*, *CEUR Workshop Proceedings*, *Communications in Computer and Information Science*, and *Education Sciences*. This distribution underscores the interdisciplinary character of the field, in which both education-focused journals and computing-oriented proceedings serve as major publication pathways.

Citation visibility presents a related but distinct picture. *Computers and Education: Artificial Intelligence* also ranks highly by total citations, indicating that it is both a high-volume and high-visibility venue for AI literacy research. However, some venues that are less prominent by publication count appear more prominently when ranked by citations, including *Conference on Human Factors in Computing Systems - Proceedings*, *International Journal of Educational Technology in Higher Education*, and *Computers and Education Open*. This suggests that publication volume and citation impact capture different dimensions of venue prominence. Some outlets function as frequent publication channels, while others contain fewer but more highly cited contributions.

Figure 1. Top Venues for AI Literacy Research

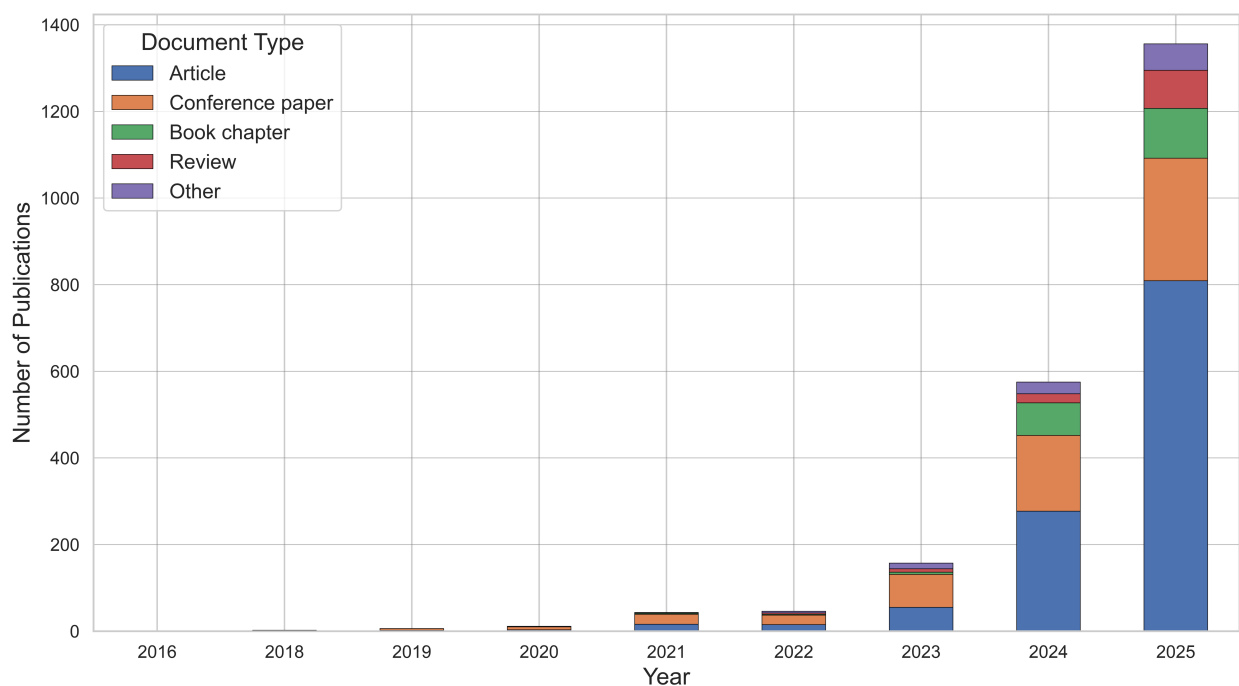


Citation-normalized venue patterns provide an additional perspective on dissemination influence. While publication counts identify the outlets where AI literacy research appears most frequently, and total citations identify venues with the largest accumulated citation visibility, mean citations per paper highlight venues where individual AI literacy articles have attracted especially strong attention. On this measure, several lower-volume venues stand out, including *Nature Human Behaviour*, *Behaviour and Information Technology*, *Smart Learning Environments*, *Mathematics*, *IEEE Transactions on Engineering Management*, and *Artificial Intelligence in Medicine*. The *International Journal of Educational Technology in Higher Education* also shows strong per-paper citation performance, suggesting that educational technology venues have contributed influential work, particularly in higher education and digital learning contexts. Because mean citations are

sensitive to small numbers of publications, however, this indicator should be interpreted as citation intensity rather than overall venue prominence.

Figure 2 shows the annual growth of AI literacy publications by document type. Publication output remained limited through 2022 but increased sharply after 2023, with especially large growth in 2024 and 2025. Articles account for the largest share of this expansion, followed by conference papers. Book chapters and reviews also became more visible in the most recent years, indicating that AI literacy research has expanded across multiple scholarly formats rather than through journal articles alone. The total height of each bar represents annual publication output, while the segments show the distribution of major document types. “Other” includes lower-frequency document types such as notes, editorials, books, and conference reviews. This pattern suggests that the rapid growth of AI literacy research has been driven primarily by formal journal publications, while conferences and edited scholarly formats remain important dissemination pathways.

Figure 2. Growth of AI Literacy Research by Document Type



Overall, the dissemination patterns show that AI literacy research is both rapidly expanding and institutionally diversifying. The field is no longer limited to a narrow set of education or computing venues; rather, it is distributed across journals, conference proceedings, book chapters, and review outlets spanning multiple disciplinary communities. At the same time, the concentration of publications and citations in a relatively small number of venues suggests that the field is beginning to develop identifiable publication hubs. These hubs are especially visible in educational technology, computer science, and human–computer interaction, reflecting the hybrid intellectual foundations of AI literacy research.

RQ2: Knowledge Production and Collaboration in AI Literacy Research

RQ2 examined patterns of knowledge production and collaboration in AI literacy research by analyzing leading authors, author publication trajectories, and co-authorship structure. Together, these analyses show that AI literacy scholarship is shaped by a relatively small group of highly productive contributors, but that author prominence varies depending on whether it is measured by publication volume, citation visibility, temporal activity, or network position.

Figure 3 presents the leading authors in AI literacy research by publication output and citation visibility. Panel A shows the ten most productive authors by number of publications, while Panel B shows total citations for the same set of authors, reordered by citation count. Ng, D.T.K. is the most productive author and also has the highest citation visibility among the top authors, indicating a prominent role in both output and scholarly influence. Chiu, T.K.F. and Breazeal, C. also appear among the most productive contributors, while Long, D. and Su, J. rank more prominently when citation visibility is considered. This difference in ordering suggests that author productivity and citation impact capture related but distinct dimensions of scholarly prominence. Some authors contribute through sustained publication activity, while others achieve higher visibility through smaller but more highly cited bodies of work.

Figure 3. Leading Authors in AI Literacy Research

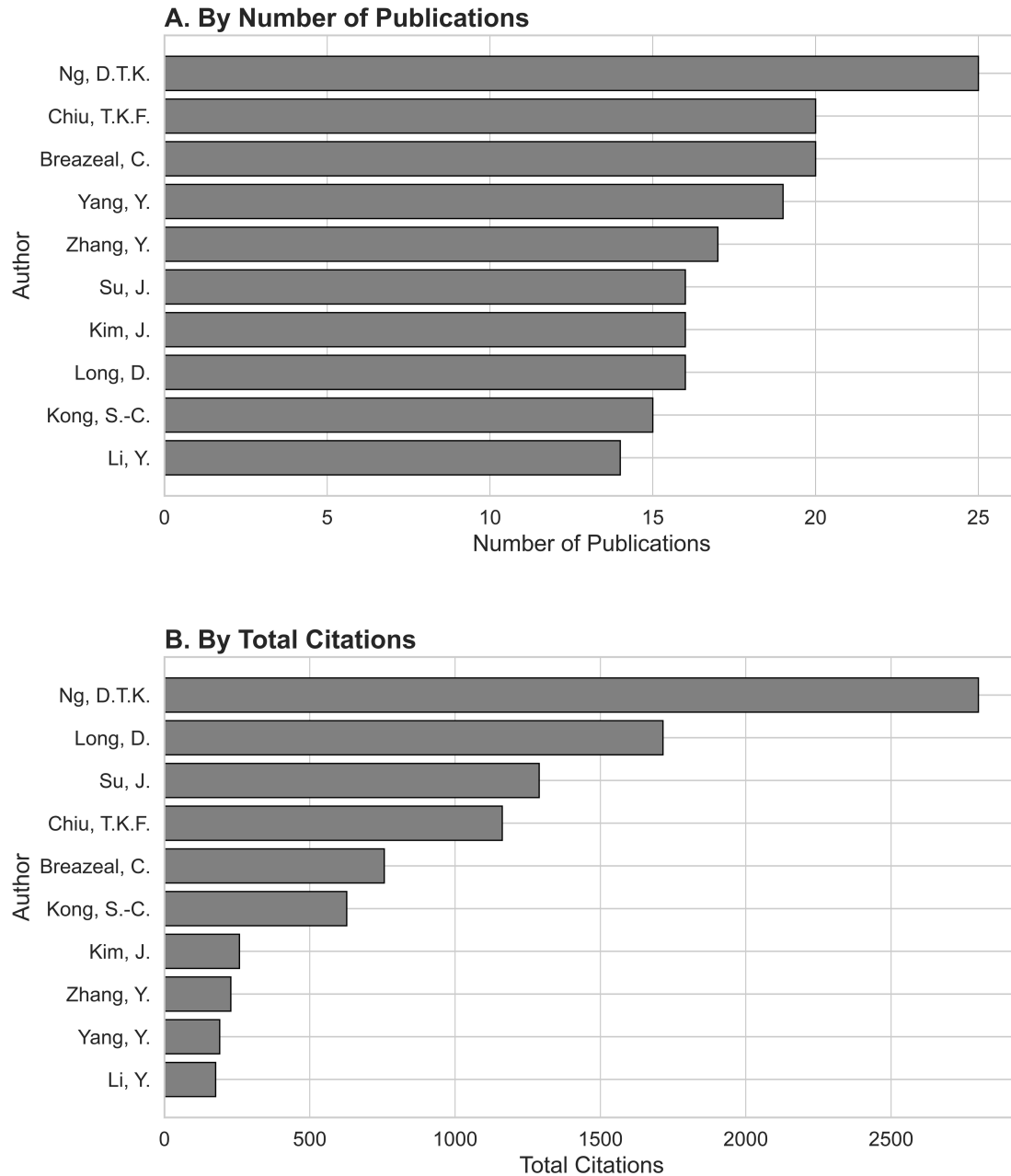


Figure 4 extends the author-level analysis by showing annual publication trajectories for the 15 most productive authors in the corpus. Bubble size represents the number of publications associated with each author in a given year, while color intensity represents the total citations accumulated by papers associated with each author-year. The figure shows that leading authors entered the AI literacy literature at different points and that productivity became especially concentrated in 2024 and 2025. Several authors, including Ng, D.T.K., Chiu, T.K.F., Breazeal, C., Yang, Y., Zhang, Y., and Kim, J., show large publication bubbles in the most recent years, reflecting the rapid acceleration of the field. At the same time, darker citation bubbles in earlier years indicate that some earlier publications have accumulated substantial citation visibility. This pattern

suggests that author prominence in AI literacy research reflects both recent productivity and the longer-term citation performance of earlier contributions.

Figure 4. Publication Trajectories of Leading Authors

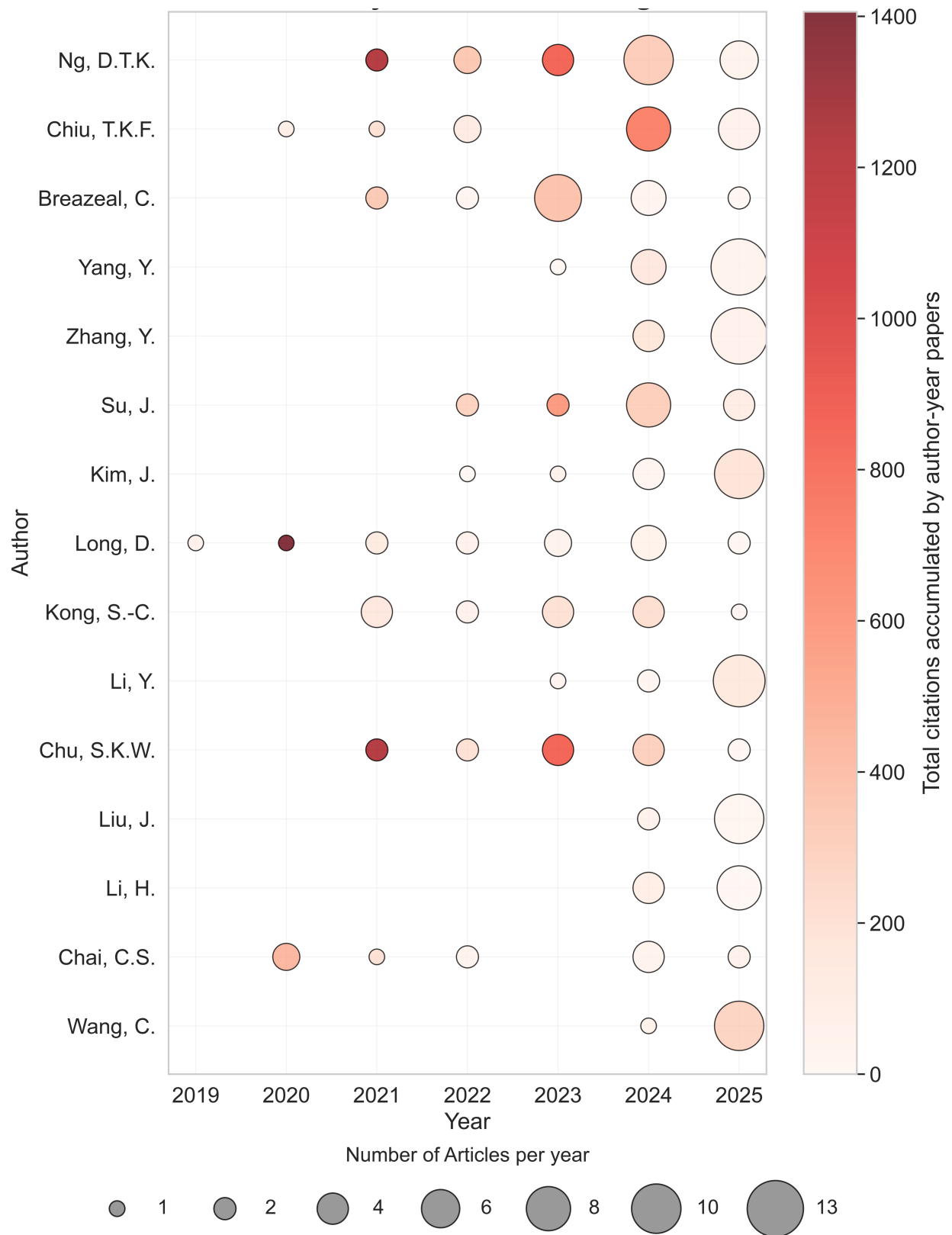
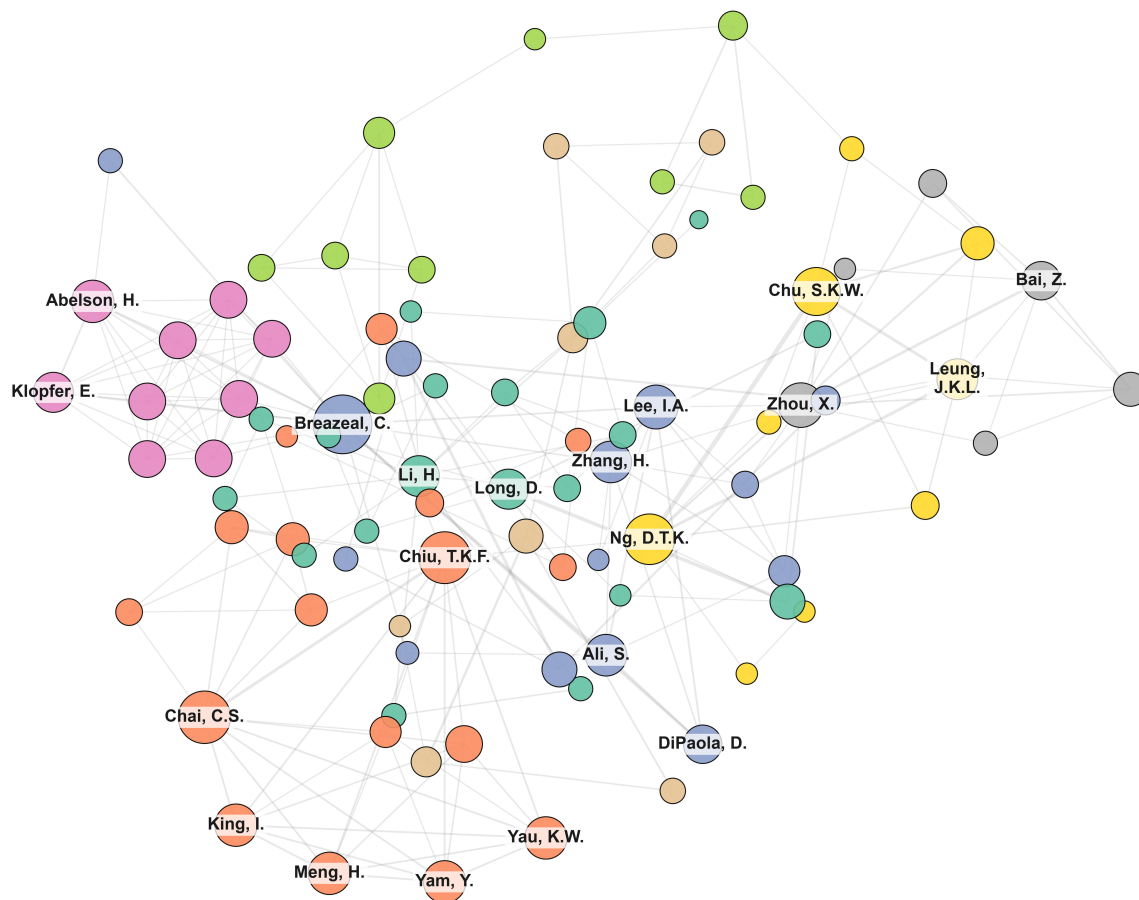


Figure 5 shows the co-authorship network of AI literacy researchers. Nodes represent authors and edges represent repeated co-authorship ties. The network includes authors connected to the 30

most productive authors through co-authored publications, with only ties appearing in at least two publications shown. Node size reflects weighted degree, edge thickness indicates the number of co-authored publications, and node colors indicate communities detected through modularity-based clustering. Labels are shown for the 20 most central authors.

The network reveals a moderately connected collaboration structure organized around several partially overlapping communities rather than a single unified core. Authors such as Chiu, T.K.F., Ng, D.T.K., Breazeal, C., Chai, C.S., Chu, S.K.W., Lee, I.A., and Zhou, X. occupy visible positions in the network, suggesting that they function as central or bridging contributors within the field. The presence of multiple-colored clusters indicates that AI literacy research is developing through distinct collaboration communities, likely reflecting differences in institutional, regional, methodological, or topical networks. At the same time, the cross-cluster ties suggest that these communities are not entirely isolated; rather, the field shows signs of partial integration through recurring collaborations among central authors.

Figure 5. Co-authorship Network of AI Literacy Researchers



Overall, the results for RQ2 indicate that knowledge production in AI literacy research is concentrated but not monolithic. A small group of authors accounts for a visible share of publication output and citation visibility, yet the temporal and network analyses show that author prominence is multidimensional. Some authors are influential because of high publication volume, others because of highly cited contributions, and others because of central positions in collaboration networks. The co-authorship structure further suggests that AI literacy research is emerging through multiple collaboration hubs rather than through a single dominant scholarly community. This pattern is consistent with the interdisciplinary character of the field, where research activity spans education, computer science, human–computer interaction, educational technology, and related domains.

RQ3: Geographic Distribution and International Collaboration in AI Literacy Research

RQ3 examined the geographic distribution of AI literacy research and the structure of international collaboration. The analysis shows that AI literacy scholarship is globally distributed but unevenly concentrated, with a small number of countries accounting for a large share of publication output and citation visibility. At the same time, international co-authorship patterns indicate that the field is not confined to isolated national research communities; rather, several major countries function as collaboration hubs linking regional and cross-regional research activity.

Figure 6 shows the global distribution of AI literacy publications by author-affiliation country. Countries are shaded according to the number of AI literacy publications associated with their author affiliations, with each country counted once per document when it appears in the affiliation information. The map indicates that AI literacy research is concentrated in a relatively small number of countries, particularly the United States and China. Additional activity is visible across parts of Europe, East and South Asia, and Oceania, suggesting that AI literacy research has developed across multiple regional contexts rather than within a single national setting. At the same time, the map also shows uneven global participation, with many countries represented by low or no publication counts in the corpus.

The five most productive countries further illustrate this concentration. The United States leads by publication volume, with 577 articles and 8,126 total citations. China ranks second with 273 articles and 2,952 citations, followed by Germany with 166 articles and 2,569 citations. Hong Kong ranks fourth by publication volume, with 159 articles, but shows exceptionally high citation visibility, with 8,125 total citations and a mean of 51.10 citations per article. The United Kingdom also appears among the leading contributors, with 156 articles and 1,470 citations. These patterns suggest that geographic prominence varies depending on whether it is measured by publication volume, total citation impact, or citation intensity. In particular, Hong Kong stands out as a high-impact contributor relative to its publication volume.

The single-country and multi-country publication indicators also reveal differences in collaboration orientation. The United States has the largest number of single-country publications (SCP = 444) and a substantial number of multi-country publications (MCP = 134), indicating both strong domestic output and international collaboration. China also shows a large domestic publication base (SCP = 177) alongside considerable international collaboration (MCP = 96). Germany and Hong Kong show more balanced patterns between single-country and multi-country output, while the United Kingdom has more multi-country publications (MCP = 95) than single-country publications (SCP = 61). This suggests that the United Kingdom plays a particularly internationalized role in AI literacy research.

Figure 6. Global Distribution of AI Literacy Publications

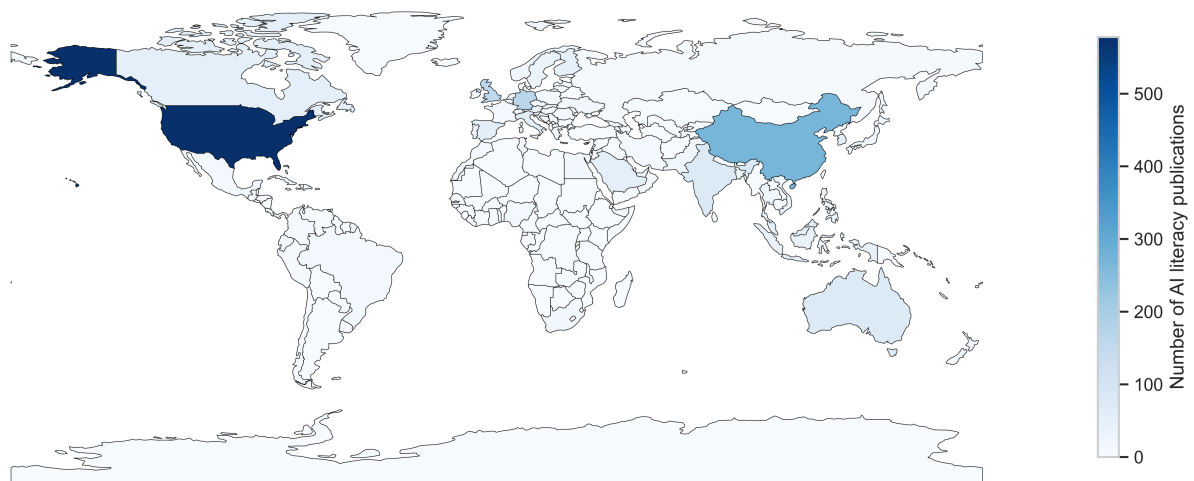
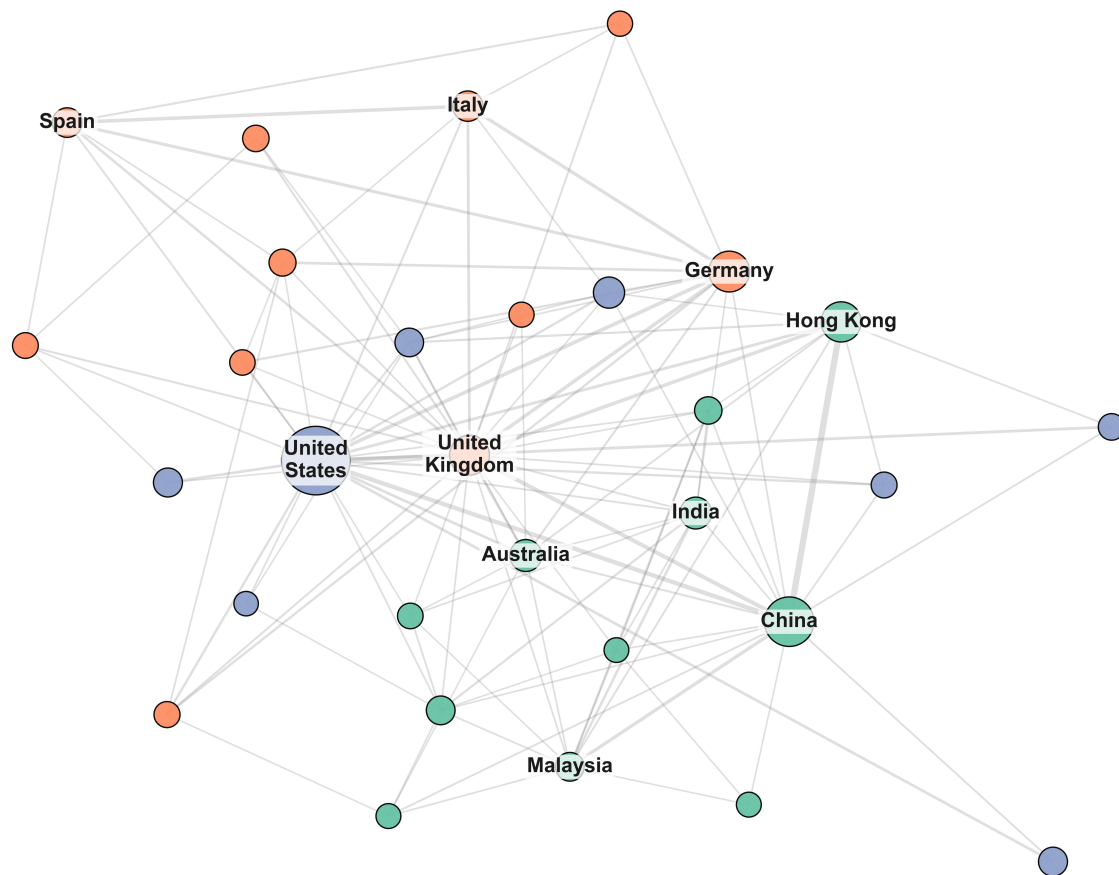


Figure 7 shows the country collaboration network in AI literacy research. Nodes represent countries and edges represent international co-authorship ties. The network includes countries with at least 20 AI literacy publications, and only country-pair collaborations appearing in at least three publications are shown. Node size reflects publication output, edge thickness indicates the number of co-authored publications between countries, and node colors indicate collaboration communities detected through modularity-based clustering. Labels are shown for the most central countries.

The network shows that international collaboration is organized around several major country hubs. The United States and the United Kingdom occupy central positions, indicating their roles as important connectors in the international collaboration structure. China and Hong Kong also appear prominently, with a strong visible collaboration area involving Asia-Pacific countries. Germany, Spain, and Italy are visible within the European collaboration structure, while Australia, India, and Malaysia appear as important contributors connecting different parts of the network. The network therefore suggests that AI literacy research is internationally connected but still organized around a limited number of highly active countries.

Figure 7. Country Collaboration Network in AI Literacy Research



Overall, figures 6 and 7 show that AI literacy research is both geographically concentrated and internationally collaborative. The map highlights the dominance of a few high-output countries, while the collaboration network shows how these countries are connected through repeated international co-authorship ties. The United States and China lead in publication volume, Hong Kong stands out for citation intensity, and the United Kingdom appears especially central in international collaboration. RQ3 indicates that AI literacy research has developed as a globally distributed but uneven field, with knowledge production concentrated in a small number of countries and collaboration structured around several major regional and cross-regional hubs. Appendix B provides full country-level statistics.

RQ4: Thematic Structure of AI Literacy Research

RQ4 examined the thematic structure of AI literacy research as a self-identified scholarly discourse. The topic model identified ten thematic concentrations, showing that AI literacy research is organized around education, generative AI, adoption and attitudes, domain-specific

professional practice, and public-facing AI systems. These topics should be interpreted as overlapping thematic areas rather than discrete subfields, reflecting the interdisciplinary and semantically connected nature of the literature.

Table 2 summarizes the ten thematic concentrations, including topic size, corpus share, and representative terms.

Table 2. Thematic Concentrations in AI Literacy Research

Topic	Manual topic label	n	% of corpus	Representative terms
0	AI literacy competencies, educator readiness, and ethical understanding	268	12.6%	understanding; educators; ethical; competencies; children; curriculum; assessment; school
1	Self-efficacy, acceptance, anxiety, and AI adoption factors	260	12.3%	self-efficacy; intention; perceived; technology; anxiety; scale; acceptance
2	Higher education integration, ethics, and professional practice	258	12.2%	ethical; higher; integration; educators; critical; ethics; challenges; academic
3	Generative AI literacy and instructional tool integration	241	11.4%	GenAI; generative; tools; critical; writing; assessment; challenges
4	School curriculum, machine learning, and STEM-oriented AI education	235	11.1%	school; design; curriculum; machine learning; course; science; concepts
5	ChatGPT, writing, language learning, and prompting	220	10.4%	ChatGPT; writing; language; critical; tools; academic; LLMs; prompt
6	Human decision-making, trust, and public-facing AI systems	188	8.9%	human; decision; systems; ethical; public; trust; users
7	Children, games, creative activities, and experiential AI learning	187	8.8%	children; design; activities; game; creative; young; generative
8	Healthcare and medical AI literacy	173	8.2%	healthcare; health; medical; care; clinical; nursing; patient
9	Libraries, librarians, and information services	90	4.2%	libraries; library; librarians; academic; information; services

***Note.** Percentages are based on the topic-modeling corpus of 2,120 records. Representative terms are derived from the topic representation and are used to support interpretation. Topics should be interpreted as overlapping thematic concentrations rather than mutually exclusive categories.*

The largest topic, *AI literacy competencies, educator readiness, and ethical understanding*, accounted for 268 documents, or 12.6% of the topic-modeling corpus. This topic centers on

foundational AI literacy competencies, teacher preparation, ethical awareness, assessment, and curriculum development. It reflects a major strand of the literature concerned with what learners and educators should understand about AI and how these competencies can be incorporated into educational practice.

The second topic, *self-efficacy, acceptance, anxiety, and AI adoption factors*, accounted for 260 documents, or 12.3% of the corpus. This topic focuses on individual attitudes toward AI, including perceived usefulness, acceptance, self-efficacy, anxiety, intention to use, and technology adoption. Its prominence suggests that AI literacy research is not limited to knowledge or skills but also examines psychological and behavioral conditions that shape whether individuals feel prepared to engage with AI systems.

The third topic, *higher education integration, ethics, and professional practice*, included 258 documents, or 12.2% of the corpus. This topic captures research on AI literacy in universities and professional learning contexts, including academic integrity, ethical use, disciplinary integration, and institutional challenges. Its size indicates that higher education has become one of the central sites in which AI literacy is being defined, taught, and contested.

The fourth topic, *generative AI literacy and instructional tool integration*, included 241 documents, or 11.4% of the corpus. This topic reflects the rapid emergence of generative AI as a major focus of AI literacy research. It includes work on GenAI tools, instructional design, assessment, critical use, and the challenges of integrating generative AI into educational settings. The prominence of this topic demonstrates how strongly recent AI literacy scholarship has been shaped by the spread of generative AI technologies.

The fifth topic, *school curriculum, machine learning, and STEM-oriented AI education*, accounted for 235 documents, or 11.1% of the corpus. This topic represents a more established strand of AI literacy research focused on school-based learning, machine learning concepts, curriculum design, computing education, and STEM-oriented instruction. It reflects the field's earlier roots in K–12 AI education and computational learning.

The sixth topic, *ChatGPT, writing, language learning, and prompting*, included 220 documents, or 10.4% of the corpus. This topic is closely related to generative AI but is more specifically focused on large language models, writing, academic communication, language learning, and prompt-based interaction. Its separation from the broader generative AI topic suggests that ChatGPT-mediated writing and prompting have become sufficiently prominent to form a distinct thematic concentration within AI literacy research.

The remaining topics represent more specialized but important areas of the field. *Human decision-making, trust, and public-facing AI systems* accounted for 188 documents, or 8.9% of the corpus, connecting AI literacy to public understanding, trust, fairness, algorithmic decision-making, and user interaction with AI systems. *Children, games, creative activities, and experiential AI learning* included 187 documents, or 8.8%, and focused on age-appropriate AI learning through creative, playful, and experiential activities. *Healthcare and medical AI literacy* accounted for 173 documents, or 8.2%, highlighting clinical practice, nursing, patient care, and health-professional readiness. Finally, *libraries, librarians, and information services* was the smallest topic, with 90

documents, or 4.2%, but it identifies an important institutional setting in which librarians and information professionals mediate AI literacy, information access, and responsible AI use.

Overall, the topic distribution shows that AI literacy research has developed into a multidimensional field. Education remains the central organizing context, but the literature now extends well beyond school curriculum and teacher preparation. AI literacy research also addresses higher education, professional practice, generative AI tools, writing and prompting, healthcare, libraries, public-facing AI systems, and user trust. This thematic breadth suggests that AI literacy is increasingly understood not as a single technical competency, but as a broad sociotechnical construct spanning knowledge, skills, ethics, attitudes, institutional practices, and domain-specific applications. Supplementary diagnostics for the 10-topic solution, including cluster-quality metrics, semantic similarity checks, topic-word overlap, stability across random seeds, and comparison with alternative topic-number solutions, are provided in Appendix C.

To examine how the field changed over time, topic prevalence was compared across three periods: publications before 2023, the transition year of 2023, and the 2024–2025 period. Because publication volume increased sharply after 2023, the analysis focuses on topic shares within each period rather than raw counts alone. This makes it possible to distinguish overall field growth from changes in relative thematic emphasis. Table 3 reports topic prevalence across the three periods and the percentage-point change from the pre-2023 period to 2024–2025.

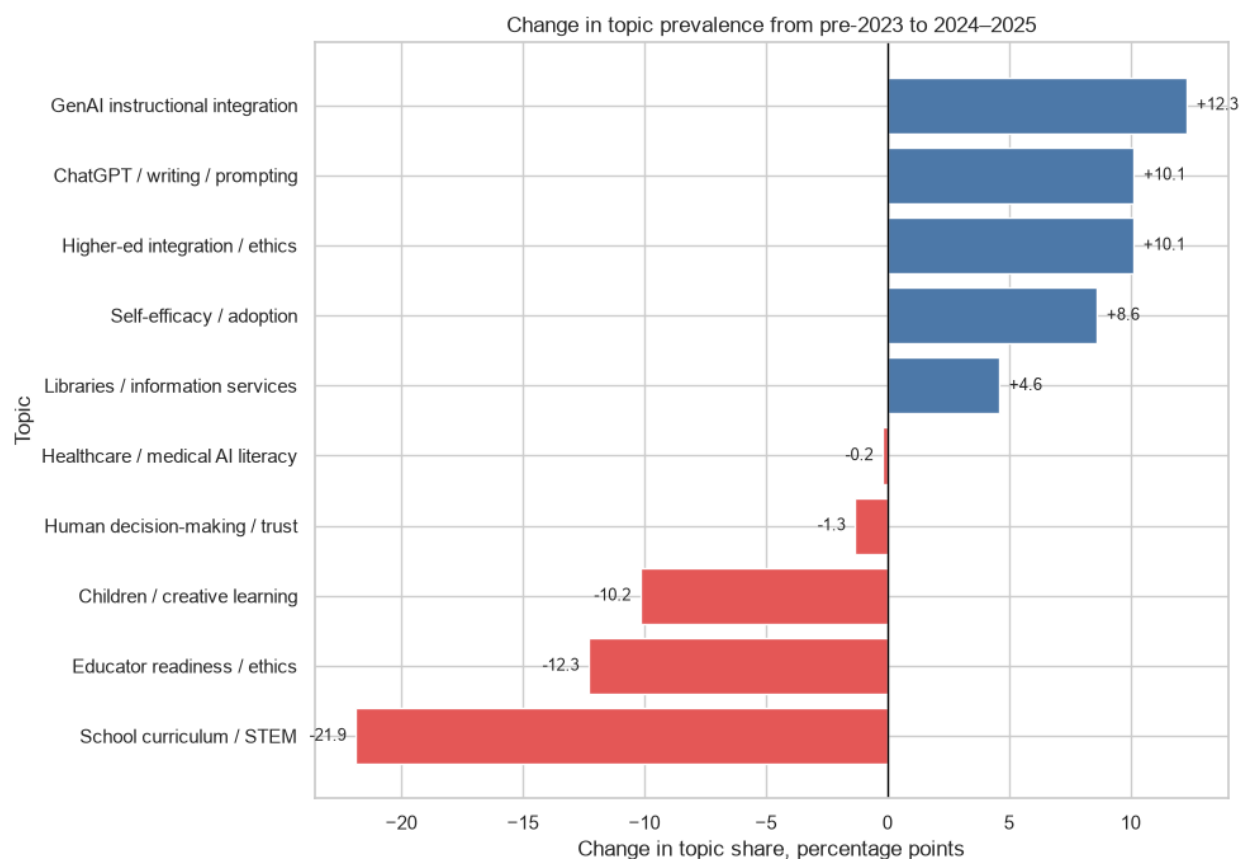
Table 3. Topic Prevalence by Period

Topic	Manual topic label	Pre-2023 %	2023 %	2024–2025 %	Change
3	Generative AI literacy and instructional tool integration	0.0	7.5	12.3	+12.3
5	ChatGPT, writing, language learning, and prompting	0.9	8.2	11.1	+10.1
2	Higher education integration, ethics, and professional practice	2.8	8.8	13.0	+10.1
1	Self-efficacy, acceptance, anxiety, and AI adoption factors	4.7	4.1	13.3	+8.6
9	Libraries, librarians, and information services	0.0	2.7	4.6	+4.6
8	Healthcare and medical AI literacy	8.5	6.1	8.3	-0.2
6	Human decision-making, trust, and public-facing AI systems	10.4	5.4	9.1	-1.3
7	Children, games, creative activities, and experiential AI learning	17.9	15.6	7.8	-10.2
0	AI literacy competencies, educator readiness, and ethical understanding	23.6	21.8	11.3	-12.3
4	School curriculum, machine learning, and STEM-oriented AI education	31.1	19.7	9.3	-21.9

***Note.** Values indicate the percentage of documents within each period assigned to each topic. Change is calculated as the percentage-point difference between the pre-2023 period and the 2024–2025 period. Positive values indicate increased relative prevalence; negative values indicate decreased relative prevalence.*

Figure 8 visualizes the same change scores, highlighting the topics that gained or declined most strongly in relative prevalence.

Figure 8: Change in Topic Prevalence from Pre-2023 to 2024–2025



The results show a clear thematic reorientation after 2023. Before 2023, AI literacy research was dominated by school-based and foundational education topics. School curriculum, machine learning, and STEM-oriented AI education accounted for 31.1% of pre-2023 documents, making it the largest early theme. AI literacy competencies, educator readiness, and ethical understanding accounted for 23.6%, while children, games, creative activities, and experiential AI learning accounted for 17.9%. These topics suggest that early AI literacy research was strongly rooted in K–12 education, curriculum design, teacher preparation, and age-appropriate learning activities.

By contrast, the 2024–2025 period shows a more diversified and generative AI-oriented thematic structure. The largest relative increase occurred in generative AI literacy and instructional tool integration, which rose from 0.0% before 2023 to 12.3% in 2024–2025. ChatGPT, writing, language learning, and prompting increased from 0.9% to 11.1%, while higher education

integration, ethics, and professional practice increased from 2.8% to 13.0%. Self-efficacy, acceptance, anxiety, and AI adoption factors also increased substantially, from 4.7% before 2023 to 13.3% in 2024–2025. These increases indicate that the rise of generative AI shifted the field toward tool use, prompting, academic work, higher education, and adoption readiness.

Several earlier themes declined in relative share. School curriculum, machine learning, and STEM-oriented AI education decreased from 31.1% before 2023 to 9.3% in 2024–2025. AI literacy competencies, educator readiness, and ethical understanding declined from 23.6% to 11.3%, and children, games, creative activities, and experiential AI learning declined from 17.9% to 7.8%. These decreases should not be interpreted as disappearance. Rather, they indicate that earlier education-centered themes became proportionally smaller as newer GenAI-related, higher education, writing, and adoption-oriented topics expanded rapidly.

Other topics showed more stable or specialized trajectories. Healthcare and medical AI literacy remained comparatively stable across periods, changing only slightly from 8.5% before 2023 to 8.3% in 2024–2025. Human decision-making, trust, and public-facing AI systems declined modestly in relative share, while libraries, librarians, and information services emerged as a smaller but newly visible topic in the most recent period. This suggests that the post-2023 expansion of AI literacy research did not only affect education, but also created new institutional roles for libraries, information professionals, healthcare educators, and other domain-specific actors.

Overall, the temporal analysis shows that AI literacy research has shifted from an earlier focus on foundational AI education, school curriculum, teacher readiness, and children’s learning toward a more differentiated field shaped by generative AI. The post-2023 literature places greater emphasis on GenAI tools, ChatGPT-mediated writing and prompting, higher education integration, user acceptance, and professional readiness. This shift indicates that the rise of generative AI did not simply increase the volume of AI literacy research; it also reorganized the field’s thematic priorities.

Discussion

This study mapped AI literacy research as a self-identified scholarly discourse by examining its dissemination patterns, collaboration structures, geographic distribution, and thematic organization. The findings suggest that AI literacy has moved from a relatively marginal topic to a rapidly expanding area of scholarship, particularly after the emergence of generative AI (Table 1; Figure 2). The significance of this growth, however, lies not only in publication volume. More importantly, the findings suggest that AI literacy is becoming a shared scholarly label through which different communities are negotiating what people need to know, do, and judge in AI-mediated environments.

Field Formation

The findings indicate that AI literacy is undergoing field formation, but not through convergence around a single disciplinary home or theoretical definition. The emergence of identifiable publication venues, collaboration hubs, and geographic centers suggests that AI literacy has acquired greater scholarly visibility (Figures 1–7). Yet this visibility remains distributed across several communities rather than organized around one dominant center. This pattern is important because it suggests that AI literacy is not simply a subtopic within AI education or digital literacy. It is becoming a broader field-level concern about human understanding, agency, and responsible participation in AI-shaped settings (Long & Magerko, 2020; Ng et al., 2021).

At the same time, the field's development remains uneven. The venue patterns show that publication volume and citation visibility do not always align (Figure 1). This suggests that different outlets are contributing to field formation in different ways: some provide high-volume publication channels, while others give visibility to fewer but more influential contributions. Rather than treating this as fragmentation alone, it may be more useful to view it as evidence of a field still assembling its intellectual infrastructure.

This helps explain why AI literacy continues to appear conceptually heterogeneous. Prior research has already noted that AI literacy includes multiple dimensions, including technical understanding, practical use, ethical awareness, and critical evaluation (Ng et al., 2021; Almatrafi et al., 2024). The present findings add that this heterogeneity is also structural. Different communities use the AI literacy label to address different institutional problems, including curriculum design, professional readiness, trust, writing, healthcare practice, and information access. The challenge for the field, therefore, is not necessarily to eliminate plurality, but to clarify how these strands relate to one another.

Generative AI and Conceptual Change

The temporal pattern in the results indicates that generative AI has reorganized the agenda of AI literacy research rather than merely increasing the number of publications (Table 3; Figure 8). Earlier work was more closely tied to foundational AI concepts, school curriculum, teacher preparation, and children's learning. More recent work shifts attention toward the situated use of AI tools, especially in higher education, writing, prompting, adoption, and professional practice. This shift is consistent with recent scholarship arguing that generative AI requires new forms of literacy related to use, evaluation, and interaction with AI-generated outputs (Walter, 2024; Ma et al., 2024).

This change expands what AI literacy is expected to mean. Earlier AI literacy work often emphasized knowing what AI is and understanding basic concepts such as machine learning, data, bias, and ethical implications. These foundations remain necessary, but they are no longer sufficient. In the generative AI context, AI literacy also involves judging when AI use is appropriate, evaluating uncertain outputs, understanding the limits of automation, and working responsibly with tools that participate in writing, reasoning, and decision-making.

The implication is that AI literacy should not be defined only as a stable body of knowledge. It also needs to include adaptive practices that change as AI systems and institutional expectations change. A useful way forward may be to distinguish between durable foundations and context-specific competencies. Durable foundations include concepts such as data, bias, opacity, ethics, and social impact. Context-specific competencies include prompting, output evaluation, disciplinary use, institutional policy awareness, and trust calibration. This distinction would allow the field to preserve conceptual continuity while still responding to technological change.

Generative AI also shifts AI literacy from an individual learning issue toward an institutional challenge. The results show that recent AI literacy research is increasingly connected to higher education, professional practice, healthcare, libraries, and public-facing AI systems (Table 2). These are not only learning contexts; they are institutional settings in which norms, policies, responsibilities, and risks must be negotiated. AI literacy, therefore, should not be framed only as something individuals lack. It should also be understood as something institutions must support through guidance, infrastructure, and responsible-use norms (Chan, 2023; Southworth et al., 2023).

Fragmentation and Visibility

The collaboration and geographic findings show that AI literacy research is internationally connected but unevenly visible (Figures 5–7). This matters because the field’s most visible knowledge producers may shape which definitions, frameworks, and problems become dominant. If AI literacy research is concentrated in particular countries, institutions, or language contexts, then the field may overrepresent some assumptions about learners, technologies, educational systems, and institutional capacity while underrepresenting others.

This uneven visibility is especially important because AI literacy is context dependent. What counts as meaningful AI literacy may differ across schools, universities, workplaces, healthcare systems, libraries, and public institutions. It may also vary across language communities, policy environments, and levels of access to AI tools. As a result, global AI literacy frameworks should be treated as starting points rather than universal templates. The field needs more research from underrepresented regions and institutional contexts to understand how AI literacy is practiced under different social, cultural, and technological conditions.

At the same time, fragmentation should not be treated only as a weakness. Multiple collaboration clusters can support specialized knowledge across domains, and the topic structure shows that AI literacy research now spans several applied settings (Table 2). The risk is not specialization itself, but disconnection. If domain-specific communities develop in isolation, the field may accumulate many local frameworks without building cumulative theory. Future research should therefore strengthen connections across sectors and regions while preserving sensitivity to local context.

Implications and Limitations

For research, the findings suggest that AI literacy should be treated as a multidimensional and evolving construct rather than a single competency. The topic structure shows that the field now includes conceptual, practical, affective, ethical, institutional, and domain-specific dimensions (Table 2). Future studies should therefore be explicit about which dimension of AI literacy they examine. This is especially important for measurement research, because a scale designed to assess conceptual knowledge may not capture prompting ability, confidence, ethical judgment, trust calibration, or professional readiness (Lintner, 2024).

For practice, the findings point toward context-sensitive AI literacy design. A single curriculum or framework is unlikely to work equally well across educational, professional, healthcare, library, and public settings. Instead, AI literacy initiatives should begin by asking what forms of AI use, risk, judgment, and responsibility matter in a given context. In schools, this may require age-appropriate conceptual foundations. In higher education, it may require guidance on generative AI use, academic integrity, and disciplinary practice. In healthcare and other professional settings, it may require attention to safety, accountability, and the limits of AI-supported decisions. In libraries and information services, it may require support for public understanding, information evaluation, and equitable access.

For institutions and policymakers, the findings support a cautious conclusion: AI literacy should not be framed only as an individual skill deficit. Responsible AI engagement also depends on institutional conditions, including access to tools, clear policies, professional development, assessment practices, and norms for acceptable use. The results do not support a universal policy prescription. Instead, they point toward adaptive approaches that connect AI literacy to the specific responsibilities and risks of different sectors.

Several limitations should be considered. The study relies on Scopus-indexed publications and a deliberately narrow search strategy focused on explicit uses of “AI literacy,” “A.I. literacy,” and “artificial intelligence literacy.” This improves conceptual precision but excludes adjacent work on AI education, data literacy, digital literacy, computational thinking, and algorithmic literacy that may address similar issues without using the AI literacy label. Scopus coverage may also underrepresent non-English and regionally published scholarship. Bibliometric and network analyses depend on metadata quality, including author names, affiliations, and country information. Finally, the topic model was based on titles and abstracts rather than full texts, so the identified topics should be interpreted as broad thematic concentrations rather than fixed boundaries. These limitations mean that the study should be understood as a structured map of self-identified AI literacy research, rather than a complete account of all scholarship relevant to AI literacy.

Conclusion

This study began from the premise that before AI literacy can be stabilized as a framework, curriculum, measurement construct, or policy priority, it must first be understood as a field of knowledge production. Rather than asking what AI literacy should mean in normative terms, the study examined how the term has been taken up, circulated, and organized within the scholarly literature. By mapping publications, venues, collaboration networks, geographic patterns, and

thematic concentrations, the study provides a field-level account of how self-identified AI literacy research has developed.

The findings show that AI literacy has become a rapidly expanding but unevenly organized research area. Its growth after the emergence of generative AI is not only a matter of increased publication volume. It also reflects a shift in what AI literacy is expected to address. Earlier work was more closely associated with foundational AI concepts, school curriculum, educator readiness, and children's learning. More recent work increasingly emphasizes generative AI use, ChatGPT-mediated writing and prompting, higher education integration, adoption factors, professional practice, healthcare, libraries, and public-facing AI systems. This shift suggests that AI literacy is expanding from a focus on understanding AI systems toward a broader concern with using, evaluating, and governing AI in situated contexts.

The study also shows that AI literacy is not developing as a single unified lineage moving toward conceptual consensus. Instead, it is an assembling field shaped by overlapping communities, venues, countries, and applied domains. This plurality should not be treated only as a weakness. It indicates that AI literacy has become a shared language for addressing the diverse ways AI systems enter education, work, information access, healthcare, and public life. At the same time, plurality becomes productive only when the field can clarify the relationships among its different strands. Without clearer distinctions, AI literacy risks becoming an expansive label that absorbs many concerns about AI without explaining how they relate to one another.

A central implication of this study is that future AI literacy research should distinguish more carefully among the dimensions of literacy being studied. Conceptual understanding, practical tool use, ethical reasoning, critical evaluation, trust calibration, confidence, institutional readiness, and domain-specific competence are related but not identical. Treating them as interchangeable may limit theory development, weaken measurement, and make interventions difficult to compare. More precise distinctions would allow researchers to build cumulative knowledge while still recognizing that AI literacy must vary across educational levels, professional sectors, institutional contexts, and cultural settings.

The study also points to the need for more globally and contextually diverse AI literacy research. The field is internationally connected, but visibility and influence remain unevenly distributed. As AI systems enter different societies and institutions under unequal conditions, AI literacy frameworks developed in highly visible research contexts should not be assumed to transfer directly everywhere. Future work should therefore examine how AI literacy is understood and practiced in underrepresented regions, multilingual settings, lower-resource institutions, and professional environments where the risks and responsibilities of AI use may differ substantially.

The broader significance of AI literacy research lies in its attention to a changing condition of social participation. As AI systems become embedded in classrooms, workplaces, healthcare settings, libraries, and public institutions, the central question is not only whether people can use AI effectively. It is also whether they can understand the forms of agency, dependence, uncertainty,

and responsibility that AI-mediated environments create. AI literacy research is valuable because it brings these issues into empirical and educational view. Its next task is to develop concepts, measures, and interventions that are rigorous enough to support cumulative scholarship, flexible enough to address changing technologies, and critical enough to account for the social conditions through which AI becomes meaningful.

REFERENCES

- Abeo, A. N. A., Armstrong, S., Scriney, M., & Goss, H. (2025). Artificial Intelligence Techniques and Health Literacy: A Systematic Review. *Mayo Clinic Proceedings: Digital Health*, 100269. <https://doi.org/10.1016/j.mcpdig.2025.100269>
- Alfarwan, A. (2025). Generative AI use in K-12 education: A systematic review. *Frontiers in Education*, 10, 1647573. <https://doi.org/10.3389/feduc.2025.1647573>
- Ali, M. Y., & Richardson, J. (2025). AI literacy guidelines and policies for academic libraries: A scoping review. *IFLA Journal*, 03400352251321192. <https://doi.org/10.1177/03400352251321192>
- Almatrafi, O., Johri, A., & Lee, H. (2024). A systematic review of AI literacy conceptualization, constructs, and implementation and assessment efforts (2019–2023). *Computers and Education Open*, 6, 1–20. <https://doi.org/10.1016/j.caeo.2024.100173>
- Ang, C. S. (2025). Developing AI literacy in healthcare education: Bridging the gap in competency assessment. *Discover Education*, 4(1), 372. <https://doi.org/10.1007/s44217-025-00812-z>
- Annapureddy, R., Fornaroli, A., & Gatica-Perez, D. (2025). Generative AI literacy: Twelve defining competencies. *Digital Government: Research and Practice*, 6(1), 1–21. <https://doi.org/10.1145/3685680>
- Archambault, S. G., Murph, N. L., & Ramachandran, S. (2025). Fostering AI Literacy in Undergraduates: A ChatGPT Workshop Case Study. *Library Trends*, 73(4), 443–475.

- Archambault, S. G., Ramachandran, S., Acosta, E., & Fu, S. (2024). Ethical dimensions of algorithmic literacy for college students: Case studies and cross-disciplinary connections. *The Journal of Academic Librarianship*, 50(3), 102865. <https://doi.org/10.1016/j.acalib.2024.102865>
- Baskara, F. (2025). Conceptualizing digital literacy for the AI era: A framework for preparing students in an AI-driven world. *Data and Metadata*, 4.
- Biagini, G. (2025). Towards an AI-literate future: A systematic literature review exploring education, ethics, and applications. *International Journal of Artificial Intelligence in Education*, 1–51. <https://doi.org/10.1007/s40593-025-00466-w>
- Blanco, M. A., Nelson, S. W., Ramesh, S., Callahan, C. E., Josephs, K. A., Jacque, B., & Baecher-Lind, L. E. (2025). Integrating artificial intelligence into medical education: A roadmap informed by a survey of faculty and students. *Medical Education Online*, 30(1), 2531177. <https://doi.org/10.1080/10872981.2025.2531177>
- Burrell, J. (2016). How the machine “thinks”: Understanding opacity in machine learning algorithms. *Big Data & Society*, 3(1). <https://doi.org/10.1177/2053951715622512>
- Casal-Otero, L., Catala, A., Fernández-Morante, C., Taboada, M., Cebreiro, B., & Barro, S. (2023). AI literacy in K-12: A systematic literature review. *International Journal of STEM Education*, 10(1), 29. <https://doi.org/10.1186/s40594-023-00418-7>
- Chan, C. K. Y. (2023). A comprehensive AI policy education framework for university teaching and learning. *International Journal of Educational Technology in Higher Education*, 20, Article 38. <https://doi.org/10.1186/s41239-023-00408-3>
- Chetty, K. (2023). AI literacy for an ageing workforce: Leveraging the experience of older workers. *OBM Geriatrics*, 7(3), 1–17. <https://doi.org/10.21926/obm.geriatr.2303243>

- Ciampa, K., Wolfe, Z. M., & Bronstein, B. (2023). ChatGPT in education: Transforming digital literacy practices. *Journal of Adolescent & Adult Literacy*, 67(3), 186–195.
- Crawford, K. (2021). *The Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Yale University Press.
- Darwin, Rusdin, D., Mukminatien, N., Suryati, N., Laksmi, E. D., & Marzuki. (2024). Critical thinking in the AI era: An exploration of EFL students' perceptions, benefits, and limitations. *Cogent Education*, 11(1), 2290342. <https://doi.org/10.1080/2331186X.2023.2290342>
- Domínguez Figaredo, D., & Stoyanovich, J. (2023). Responsible AI literacy: A stakeholder-first approach. *Big Data & Society*, 10(2), 1–15. <https://doi.org/10.1177/20539517231219958>
- Donthu, N., Kumar, S., Mukherjee, D., Pandey, N., & Lim, W. M. (2021). How to conduct a bibliometric analysis: An overview and guidelines. *Journal of Business Research*, 133, 285–296. <https://doi.org/10.1016/j.jbusres.2021.04.070>
- Druga, S., Vu, S. T., Likhith, E., & Qiu, T. (2019). Inclusive AI literacy for kids around the world. In *Proceedings of FabLearn 2019* (pp. 104–111). <https://doi.org/10.1145/3311890.3311904>
- Durak, G., Çankaya, S., Özdemir, D., & Can, S. (2024). Artificial intelligence in education: A bibliometric study on its role in transforming teaching and learning. *International Review of Research in Open and Distributed Learning*, 25(3), 219–244.
- Eguchi, A., Okada, H., & Muto, Y. (2021). Contextualizing AI education for K-12 students to enhance their learning of AI literacy through culturally responsive approaches. *KI–Künstliche Intelligenz*, 35(2), 153–161. <https://doi.org/10.1007/s13218-021-00737-3>

- El Arab, R. A., Al Moosa, O. A., Abuadas, F. H., & Somerville, J. (2025). The role of AI in nursing education and practice: Umbrella review. *Journal of Medical Internet Research*, 27, e69881. <https://doi.org/10.2196/69881>
- El Arab, R. A., Alshakihs, A. H., Alabdulwahab, S. H., Almubarak, Y. S., Alkhalifah, S. S., Abdrbo, A., ... & Sagbakken, M. (2025). Artificial intelligence in nursing: A systematic review of attitudes, literacy, readiness, and adoption intentions among nursing students and practicing nurses. *Frontiers in Digital Health*, 7, 1666005. <https://doi.org/10.3389/fdgth.2025.1666005>
- Floridi, L., & Chiriatti, M. (2020). GPT-3: Its nature, scope, limits, and consequences. *Minds and Machines*, 30(4), 681–694. <https://doi.org/10.1007/s11023-020-09548-1>
- Fui-Hoon Nah, F., Zheng, R., Cai, J., Siau, K., & Chen, L. (2023). Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration. *Journal of Information Technology Case and Application Research*, 25(3), 277–304. <https://doi.org/10.1080/15228053.2023.2233814>
- Gazquez-Garcia, J., Sánchez-Bocanegra, C. L., & Sevillano, J. L. (2025). AI in the health sector: Systematic review of key skills for future health professionals. *JMIR Medical Education*, 11(1), e58161. <https://doi.org/10.2196/58161>
- Giustini, D., & Dastyar, V. (2024). Critical AI literacy for interpreting in the age of AI. *Interpreting and Society*, 4(2), 196–213. <https://doi.org/10.1177/27523810241247259>
- Gu, X., & Ericson, B. J. (2025). AI literacy in K-12 and higher education in the wake of generative AI: An integrative review. In *Proceedings of the 2025 ACM Conference on International Computing Education Research* V. 1 (pp. 125–140). <https://doi.org/10.1145/3702652.3744217>

- Hossain, Z., Biswas, M. S., Khan, N. A., & Khan, G. (2025). Artificial intelligence literacy among South Asian library and information science students: Socio-demographic influences and educational implications. *IFLA Journal*, 51(3), 629–646. <https://doi.org/10.1177/03400352251331468>
- Jantakun, K., Jantakun, T., & Jantakoon, T. (2025). Bibliometric analysis of artificial intelligence for digital literacy. *Journal of Education and Learning*, 14(3), 115–128. <https://doi.org/10.5539/jel.v14n3p115>
- Kang, E. Y. N., Chen, D. R., & Chen, Y. Y. (2023). Associations between literacy and attitudes toward artificial intelligence–assisted medical consultations: The mediating role of perceived distrust and efficiency of artificial intelligence. *Computers in Human Behavior*, 139, 107529. <https://doi.org/10.1016/j.chb.2022.107529>
- Kim, S., Jang, Y., Kim, W., Choi, S., Jung, H., Kim, S., & Kim, H. (2021). Why and what to teach: AI curriculum for elementary school. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(17), 15569–15576. <https://doi.org/10.1609/aaai.v35i17.17833>
- Kimiafar, K., Sarbaz, M., Tabatabaei, S. M., Ghaddaripouri, K., Mousavi, A. S., Mehneh, M. R., & Baigi, S. F. M. (2023). Artificial intelligence literacy among healthcare professionals and students: A systematic review. *Frontiers in Health Informatics*, 12(0), 168.
- Kong, S. C., Cheung, W. M. Y., & Zhang, G. (2021). Evaluation of an artificial intelligence literacy course for university students with diverse study backgrounds. *Computers and Education: Artificial Intelligence*, 2, 100026. <https://doi.org/10.1016/j.caeai.2021.100026>
- Krakowski, S. (2025). Human-AI agency in the age of generative AI. *Information and Organization*, 35(1), 100560. <https://doi.org/10.1016/j.infoandorg.2025.100560>

- LaFlamme, K. A. (2025). Scaffolding AI literacy: An instructional model for academic librarianship. *The Journal of Academic Librarianship*, 51(3), 103041. <https://doi.org/10.1016/j.acalib.2025.103041>
- Lee, D., Arnold, M., Srivastava, A., Plastow, K., Strelan, P., Ploeckl, F., Lekkas, D., & Palmer, E. (2024). The impact of generative AI on higher education learning and teaching: A study of educators' perspectives. *Computers and Education: Artificial Intelligence*, 6, 1–10.
- Li, H., & Kim, S. (2024). Developing AI literacy in HRD: Competencies, approaches, and implications. *Human Resource Development International*, 27(3), 345–366.
- Lin, Z., Dai, Y., & Ng, O. L. (2025). Constructionism in K-12 AI literacy education: A systematic review of pedagogical designs, student outcomes, and learning mechanisms. *Journal of Educational Computing Research*, 07356331251360442. <https://doi.org/10.1177/07356331251360442>
- Lintner, T. (2024). A systematic review of AI literacy scales. *npj Science of Learning*, 9(1), 50. <https://doi.org/10.1038/s41539-024-00264-4>
- Liu, X., Zhang, L., & Wei, X. (2025). Generative artificial intelligence literacy: Scale development and its effect on job performance. *Behavioral Sciences*, 15(6), 811. <https://doi.org/10.3390/bs15060811>
- Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–16). <https://doi.org/10.1145/3313831.3376727>
- López-Pernas, S., Saqr, M., & Apiola, M. (2023). Scientometrics: A concise introduction and a detailed methodology for mapping the scientific field of computing education research. In

- Past, Present and Future of Computing Education Research: A Global Perspective (pp. 79–99).
- Luong, J., Tzang, C. C., McWatt, S., Brassett, C., Stearns, D., Sagoo, M. G., ... & Wu, A. (2025). Exploring artificial intelligence readiness in medical students: Analysis of a global survey. *Medical Science Educator*, 35(1), 331–341.
- Ma, M., Ng, D. T. K., Liu, Z., & Wong, G. K. (2025). Fostering responsible AI literacy: A systematic review of K-12 AI ethics education. *Computers and Education: Artificial Intelligence*, 100422. <https://doi.org/10.1016/j.caeai.2025.100422>
- Ma, Q., Crosthwaite, P., Sun, D., & Zou, D. (2024). Exploring ChatGPT literacy in language education: A global perspective and comprehensive approach. *Computers and Education: Artificial Intelligence*, 7, 100278. <https://doi.org/10.1016/j.caeai.2024.100278>
- McCarthy, J. (2007). From here to human-level AI. *Artificial Intelligence*, 171(18), 1174–1182. <https://doi.org/10.1016/j.artint.2007.10.009>
- Miao, J., Thongprayoon, C., Kashani, K. B., & Cheungpasitporn, W. (2025). Artificial intelligence as a tool for improving health literacy in kidney care. *PLOS Digital Health*, 4(2), e0000746. <https://doi.org/10.1371/journal.pdig.0000746>
- Michel-Villarreal, R., Vilalta-Perdomo, E., Salinas-Navarro, D. E., Thierry-Aguilera, R., & Gerardou, F. S. (2023). Challenges and opportunities of generative AI for higher education as explained by ChatGPT. *Education Sciences*, 13(9), 856. <https://doi.org/10.3390/educsci13090856>
- Miltenoff, P. (2024). Academic librarians in times of AI and AI literacy: Tasks, responsibilities, and leadership. *International Information & Library Review*, 56(3), 296–305.

- Monyela, M., & Tella, A. (2024). Leveraging artificial intelligence for sustainable knowledge organisation in academic libraries. *South African Journal of Libraries and Information Science*, 90(2), 1–11.
- Namdas, V. (2024). Role of artificial intelligence in information literacy: Enhancing digital literacy skills. *International e-Journal of Library Science*, 12(1), 1102–1107.
- Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., & Qiao, M. S. (2021). Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, 1–11. <https://doi.org/10.1016/j.caeai.2021.100041>
- Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., & Qiao, M. S. (2021). Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, Article 100041. <https://doi.org/10.1016/j.caeai.2021.100041>
- Ng, D. T. K., Leung, J. K. L., O'Dea, X. C., & Chu, S. K. W. (2026c). Conclusion: Navigating the future of generative AI literacy. In *From AI Literacy to Generative AI Literacy: Theory, Policy and Practice* (pp. 225–231). Springer Nature Singapore.
- Ng, D. T. K., Leung, J. K. L., O'Dea, X., & Chu, S. K. W. (2026a). Questionnaires for measuring AI literacy and related dimensions. In *From AI Literacy to Generative AI Literacy: Theory, Policy and Practice* (pp. 207–223). Springer Nature Singapore.
- Ng, D. T. K., Leung, J. K. L., O'Dea, X., & Chu, S. K. W. (2026b). From AI literacy to generative AI literacy: Conceptualization and frameworks. In *From AI Literacy to Generative AI Literacy: Theory, Policy and Practice* (pp. 21–38). Springer Nature Singapore.
- Ngoveni, M. (2025). Bridging the AI knowledge gap: The urgent need for AI literacy and institutional support. *The International Journal of Technologies in Learning*, 32(2), 83.

- Nordström, M. (2022). AI under great uncertainty: Implications and decision strategies for public policy. *AI & Society*, 37(4), 1703–1714. <https://doi.org/10.1007/s00146-021-01263-4>
- Oschinski, M., Crawford, A., & Wu, M. (2024). AI and the future of workforce training. Center for Security and Emerging Technology.
- Pasquale, F. (2020). *New Laws of Robotics: Defending Human Expertise in the Age of AI*. Harvard University Press.
- Pinski, M., & Benlian, A. (2024). AI literacy for users: A comprehensive review and future research directions of learning methods, components, and effects. *Computers in Human Behavior: Artificial Humans*, 2(1), 100062. <https://doi.org/10.1016/j.chbah.2024.100062>
- Polomoshnov, J., Masso, A., & Lobanova, K. (2025). Comparing digital, data, and AI literacy: A narrative review. *Information Polity*, 15701255251401863. <https://doi.org/10.1177/15701255251401863>
- Porter, A., & Foronda, C. (2024). Enhancing artificial intelligence literacy in nursing education to combat embedded biases. *Nursing Education Perspectives*, 45(2), 131–132.
- Pramod, D. (2025). Decoding responsible AI use: The influence of digital literacy and ethical awareness. *Cogent Education*, 12(1), 2592371. <https://doi.org/10.1080/2331186X.2025.2592371>
- Pu, L., Liang, J., Wang, J., Zhang, N., & Zhong, W. (2024). Impact of AI-based learning, digital literacy, information stewardship on learning outcomes. *Profesional de la Información*, 33(5), e330515. <https://doi.org/10.3145/epi.2024.ene.0515>
- Saddhono, K., Suhita, R., Rakhmawati, A., Rohmadi, M., & Sukmono, I. K. (2024). AI and literacy: Developing critical thinking and analytical skills in the digital era. In 2024

- International Conference on IoT, Communication and Automation Technology (ICICAT) (pp. 360–365). IEEE.
- Sadowski, J. (2020). *Too Smart: How Digital Capitalism is Extracting Data, Controlling Our Lives, and Taking Over the World*. MIT Press.
- Schmarzo, B. (2023). *AI & Data Literacy: Empowering Citizens of Data Science*. Packt Publishing Ltd.
- Simms, R. C. (2025). Generative artificial intelligence (AI) literacy in nursing education: A crucial call to action. *Nurse Education Today*, 146, 106544. <https://doi.org/10.1016/j.nedt.2025.106544>
- Soto-Sanfiel, M. T., Angulo-Brunet, A., & Lutz, C. (2025). The scale of artificial intelligence literacy for all (SAIL4ALL): Assessing knowledge of artificial intelligence in all adult populations. *Humanities and Social Sciences Communications*, 12(1), 1–14.
- Southworth, J., Migliaccio, K., Glover, J., Glover, J. N., Reed, D., McCarty, C., ... & Thomas, A. (2023). Developing a model for AI across the curriculum: Transforming the higher education landscape via innovation in AI literacy. *Computers and Education: Artificial Intelligence*, 4, 100127. <https://doi.org/10.1016/j.caeai.2023.100127>
- Southworth, J., Migliaccio, K., Glover, J., Glover, J., Reed, D., McCarty, C., Brendemuhl, J., & Thomas, A. (2023). Developing a model for AI across the curriculum: Transforming the higher education landscape via innovation in AI literacy. *Computers and Education: Artificial Intelligence*, 4, Article 100127. <https://doi.org/10.1016/j.caeai.2023.100127>
- Stamboliev, E. (2023). Proposing a postcritical AI literacy: Why we should worry less about algorithmic transparency and more about citizen empowerment. *Media Theory*, 7(1), 202–232.

- Stolpe, K., & Hallström, J. (2024). Artificial intelligence literacy for technology education. *Computers and Education Open*, 6, 100159. <https://doi.org/10.1016/j.cao.2024.100159>
- Subaveerapandiyan, A., Mvula, D., Ahmad, N., Taj, A., & Ahmed, M. G. (2024). Assessing AI literacy and attitudes among medical students: Implications for integration into healthcare practice. *Journal of Health Organization and Management*.
- Tan, Q., & Tang, X. (2025). Unveiling AI literacy in K-12 education: A systematic literature review of empirical research. *Interactive Learning Environments*, 1–17.
- Tang, K. S., Cooper, G., Rappa, N., & Edwards, J. (2025). Critical questioning with generative AI: Developing AI literacy in secondary education. *Thinking Skills and Creativity*, 102043. <https://doi.org/10.1016/j.tsc.2025.102043>
- Tenório, K., Olari, V., Chikobava, M., & Romeike, R. (2023). Artificial intelligence literacy research field: A bibliometric analysis from 1989 to 2021. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1* (pp. 1083–1089). <https://doi.org/10.1145/3545945.3569826>
- Thierer, A. D., Castillo O'Sullivan, A., & Russell, R. (2017). Artificial intelligence and public policy. *Mercatus Research Paper*.
- Tiernan, P., Costello, E., Donlon, E., Parysz, M., & Scriney, M. (2023). Information and media literacy in the age of AI: Options for the future. *Education Sciences*, 13(9), 906. <https://doi.org/10.3390/educsci13090906>
- Tzanova, S. (2024). AI in academic libraries: Success, pitfalls, perceptions, and why we need AI literacy. In *Applications of Artificial Intelligence in Libraries* (pp. 19–44). IGI Global Scientific Publishing.

- Valle-Cruz, D., Alejandro Ruvalcaba-Gomez, E., Sandoval-Almazan, R., & Ignacio Criado, J. (2019). A review of artificial intelligence in government and its potential from a public policy perspective. In Proceedings of the 20th Annual International Conference on Digital Government Research (pp. 91–99). <https://doi.org/10.1145/3325112.3325242>
- Van Brummelen, J., Heng, T., & Tabunshchyk, V. (2021). Teaching tech to talk: K-12 conversational artificial intelligence literacy curriculum and development tools. Proceedings of the AAAI Conference on Artificial Intelligence, 35(17), 15655–15663. <https://doi.org/10.1609/aaai.v35i17.17844>
- Velander, J., Otero, N., & Milrad, M. (2024). What is critical (about) AI literacy? Exploring conceptualizations present in AI literacy discourse. In Framing Futures in Postdigital Education: Critical Concepts for Data-Driven Practices (pp. 139–160). Springer Nature Switzerland.
- Walter, Y. (2024). Embracing the future of artificial intelligence in the classroom: The relevance of AI literacy, prompt engineering, and critical thinking in modern education. International Journal of Educational Technology in Higher Education, 21(1), 15. <https://doi.org/10.1186/s41239-024-00448-3>
- Wang, K., Cui, W., & Yuan, X. (2025). Artificial intelligence in higher education: The impact of need satisfaction on artificial intelligence literacy mediated by self-regulated learning strategies. Behavioral Sciences, 15(2), 165. <https://doi.org/10.3390/bs15020165>
- Wang, N., & Lester, J. (2023). K-12 education in the age of AI: A call to action for K-12 AI literacy. International Journal of Artificial Intelligence in Education, 33(2), 228–232. <https://doi.org/10.1007/s40593-023-00358-x>

- Wang, S., Wang, F., Zhu, Z., Wang, J., Tran, T., & Du, Z. (2024). Artificial intelligence in education: A systematic literature review. *Expert Systems with Applications*, 252, 124167. <https://doi.org/10.1016/j.eswa.2024.124167>
- Wilton, L., Ip, S., Sharma, M., & Fan, F. (2022). Where is the AI? AI literacy for educators. In *International Conference on Artificial Intelligence in Education* (pp. 180–188). Springer.
- Xi, B., Hussin, N., & Kamarudin, H. D. (2025). An investigation of artificial intelligence (AI) literacy competency for library and information science (LIS) students in China. *Journal of Information and Knowledge Management*, 15(SI2), 113–122.
- Yang, Y., Zhang, Y., Sun, D., He, W., & Wei, Y. (2025). Navigating the landscape of AI literacy education: Insights from a decade of research (2014–2024). *Humanities and Social Sciences Communications*, 12(1), 1–12.
- Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education: Where are the educators? *International Journal of Educational Technology in Higher Education*, 16(1), 1–27. <https://doi.org/10.1186/s41239-019-0171-0>
- Zhang, H., Lee, I., & Moore, K. (2024). An effectiveness study of teacher-led AI literacy curriculum in K-12 classrooms. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(21), 23318–23325. <https://doi.org/10.1609/aaai.v38i21.30343>
- Zhang, H., Lee, I., Ali, S., DiPaola, D., Cheng, Y., & Breazeal, C. (2023). Integrating ethics and career futures with technical learning to promote AI literacy for middle school students: An exploratory study. *International Journal of Artificial Intelligence in Education*, 33(2), 290–324. <https://doi.org/10.1007/s40593-022-00293-3>

Zhang, K., & Aslan, A. B. (2021). AI technologies for education: Recent research and future directions. *Computers and Education: Artificial Intelligence*, 2, 100025.

<https://doi.org/10.1016/j.caeai.2021.100025>

Zhang, S., Ganapathy Prasad, P., & Schroeder, N. L. (2025). Learning about AI: A systematic review of reviews on AI literacy. *Journal of Educational Computing Research*, 07356331251342081. <https://doi.org/10.1177/07356331251342081>

Zhang, S., Ganapathy Prasad, P., & Schroeder, N. L. (2025). Learning about AI: A systematic review of reviews on AI literacy. *Journal of Educational Computing Research*, 07356331251342081. <https://doi.org/10.1177/07356331251342081>

APPENDICES

Appendix A: Scopus Search Query

This Scopus advanced search query was designed to retrieve the full scholarly output related to “AI literacy” for a comprehensive bibliometric analysis. The strategy targets occurrences of the term across conference names, source titles, and article titles, abstracts, and keywords, using several common variations (“AI literacy,” “A.I. literacy,” and “artificial intelligence literacy”). The search was limited to publications from 2010-2025 to capture the contemporary emergence and rapid growth of the field. As of November 2025, the initial Scopus search returned 2,224 records. After removing records with incomplete or unsuitable bibliographic metadata, the final bibliometric corpus contained 2,198 documents. A further exclusion of records without usable title–abstract text produced a topic-modeling corpus of 2,120 documents.

```
(CONFNAM("AI literacy")
OR CONFNAM("artificial intelligence literacy")
OR SRCTITLE("AI literacy")
OR SRCTITLE("artificial intelligence literacy")
OR TITLE-ABS-KEY("AI literacy")
OR TITLE-ABS-KEY("A.I. literacy")
OR TITLE-ABS-KEY("artificial intelligence literacy"))
AND PUBYEAR > 2009 AND PUBYEAR < 2026
```

Appendix B: Detailed Country Performance Statistics

The detailed country performance summarizes publication volume, citation impact, and collaboration structure. The United States not only produces the most publications but also shows a high number of single-country publications (SCP = 444) paired with substantial international collaborations (MCP = 134). Hong Kong stands out with an exceptionally high mean citation rate (≈ 51 citations per article), indicating strong global influence relative to its publication volume. European countries, such as Germany, Finland, Switzerland, and Sweden, also show strong citation impact, suggesting conceptual and methodological leadership. Countries like Malaysia, Saudi Arabia, and the United Arab Emirates show increasing collaboration and participation, particularly through multi-country contributions (MCP). These metrics indicate a geographically concentrated yet globally expanding field, characterized by high-impact contributions from core countries and growing engagement across diverse regions.

<i>Country</i>	<i>Number of Articles</i>	<i>Average Citations</i>	<i>Total Citations</i>	<i>SCP</i>	<i>MCP</i>
<i>United States</i>	577	14.06	8126	444	134
<i>China</i>	273	10.81	2952	177	96
<i>Germany</i>	166	15.48	2569	100	66

<i>Hong Kong</i>	159	51.1	8125	94	65
<i>United Kingdom</i>	156	9.42	1470	61	95
<i>Australia</i>	79	17.09	1350	37	42
<i>India</i>	78	10.67	832	46	32
<i>Turkey</i>	72	5.89	424	53	19
<i>Italy</i>	65	7.97	518	32	33
<i>Spain</i>	59	12.19	719	29	30
<i>Canada</i>	54	20.07	1084	34	20
<i>South Korea</i>	53	12.26	662	41	13
<i>Malaysia</i>	52	6.88	358	8	44
<i>Saudi Arabia</i>	52	9.6	499	17	35
<i>Finland</i>	51	13.2	673	22	29
<i>Indonesia</i>	42	6.12	257	24	18
<i>Austria</i>	39	15.64	610	20	19
<i>Taiwan</i>	36	16.17	582	19	17
<i>Sweden</i>	35	23.17	811	21	14
<i>Netherlands</i>	32	8.59	275	11	21
<i>Singapore</i>	31	27.16	842	14	17
<i>Portugal</i>	30	4.77	143	10	20
<i>United Arab Emirates</i>	29	2.14	62	12	17
<i>Switzerland</i>	28	24.32	681	15	13
<i>Greece</i>	26	7.42	193	15	11
<i>Viet Nam</i>	25	3.72	93	19	6
<i>Philippines</i>	24	4.79	115	18	6
<i>Pakistan</i>	24	3.08	74	6	18
<i>Thailand</i>	23	6.65	153	9	14
<i>Norway</i>	21	2.14	45	7	14

Appendix C. Topic Modeling Diagnostics and Supplementary Validation

This appendix provides supplementary diagnostic information for the topic-modeling analysis reported in the main text. The main article describes the topic-modeling corpus, preprocessing approach, KMeans-guided BERTopic workflow, model-selection rationale, and final ten-topic solution. This appendix reports additional diagnostics used to evaluate topic-size balance, embedding-space cluster quality, semantic similarity, top-word overlap, and stability across random seeds. These diagnostics were used to support the interpretation of the model as a field-level map of thematic concentrations, not as a definitive taxonomy of mutually exclusive subfields.

C.1 Topic-Size Balance

The final 10-topic solution produced a relatively balanced distribution of documents across topics. The largest topic accounted for 12.64% of the topic-modeling corpus, while the smallest accounted for 4.25%. The ratio of the largest to smallest topic was 2.98, indicating that the solution avoided both a single dominant topic and extremely small residual topics.

Diagnostic	Value
Largest topic share	0.1264
Smallest topic share	0.0425
Largest/smallest topic ratio	2.98

These diagnostics supported the use of the 10-topic solution for field-level interpretation. Although the topics varied in size, the distribution was sufficiently balanced to allow each topic to represent a meaningful thematic concentration within the corpus.

C.2 Embedding-Space Cluster Quality

Cluster-quality diagnostics were computed to assess the embedding-space structure of the final model. These metrics were interpreted comparatively rather than as absolute indicators of model quality, because topic models based on short academic texts and high-dimensional semantic embeddings often produce modest silhouette values. The metrics were therefore considered alongside interpretability, topic-size balance, redundancy, and stability across random seeds.

Metric	Value
Silhouette score, cosine	0.0524
Silhouette score, Euclidean	0.0287
Davies-Bouldin score	3.713
Calinski-Harabasz score	49.5917

The positive but low silhouette values indicate limited geometric separation among clusters. This was expected because the corpus consists of closely related AI literacy publications that often share educational, ethical, technical, and generative AI vocabularies. For this reason, the cluster-quality metrics were not treated as evidence of sharply bounded topics. Instead, they were used as comparative diagnostics to evaluate alternative topic-number solutions.

C.3 Comparison of Alternative Topic-Number Solutions

Several candidate topic solutions were evaluated to determine an appropriate number of topics for the field-level map. Table C1 reports diagnostic metrics for 8-, 10-, 12-, 14-, 15-, and 18-topic solutions. These metrics were used alongside manual inspection of topic labels, representative terms, representative documents, and redundancy among topics.

Table C1. Comparison of Alternative Topic Solutions

k	Cosine silhouette	Davies-Bouldin	Calinski-Harabasz	Largest topic share	Smallest topic share	Largest/smallest ratio
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8	0.0760	3.6768	56.8215	0.2500	0.0561	4.4538
10	0.0513	3.7085	49.6599	0.1241	0.0425	2.9222
12	0.0485	3.7444	43.9310	0.1236	0.0458	2.7010
14	0.0426	3.7847	40.0390	0.0991	0.0382	2.5926
15	0.0435	3.6538	38.1293	0.0915	0.0368	2.4872
18	0.0479	3.6477	34.0039	0.0873	0.0302	2.8906

Although the 8-topic solution had the highest cosine silhouette score, it produced a comparatively large dominant topic, suggesting that substantively distinct areas of AI literacy research were being compressed into broader clusters. Higher-topic solutions provided greater granularity but introduced additional interpretive redundancy, especially among education, generative AI, ChatGPT, and higher education topics. The 10-topic solution was selected because it provided the best balance between diagnostic performance, topic-size balance, parsimony, and substantive interpretability.

C.4 Topic Similarity Diagnostics

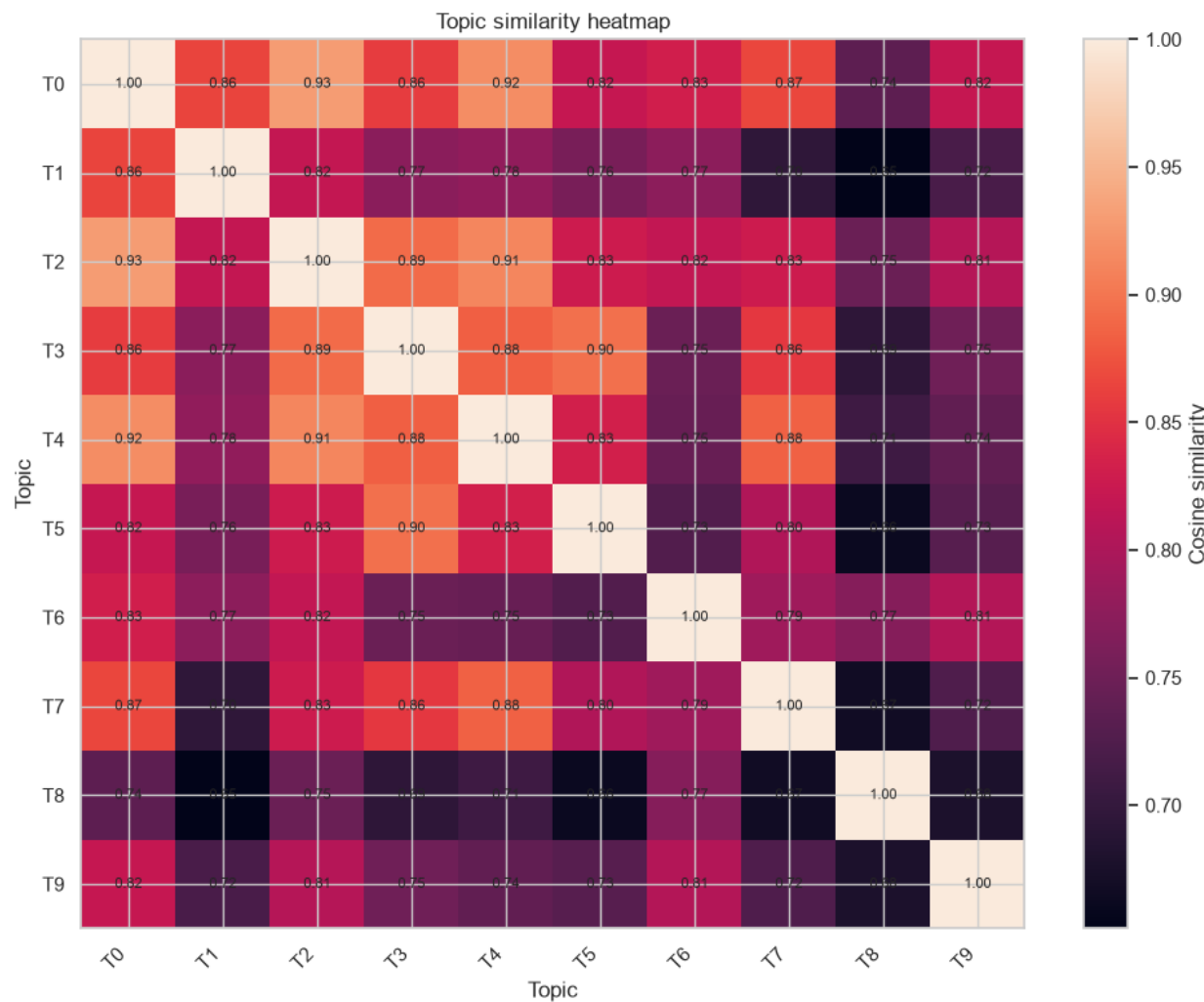
Topic-centroid similarities were examined to assess semantic proximity among topics. Several topic pairs showed high similarity, particularly those involving education, ethics, generative AI, and writing.

Topic pair	Centroid cosine similarity	Interpretation
Topic 0 – Topic 2	0.9305	Educator readiness/ethics and higher education integration/ethics
Topic 0 – Topic 4	0.9182	Educator readiness/ethics and school curriculum/STEM education
Topic 2 – Topic 4	0.9127	Higher education integration and school curriculum/STEM education
Topic 3 – Topic 5	0.8961	Generative AI instructional integration and ChatGPT/writing/prompting
Topic 2 – Topic 3	0.8914	Higher education integration and generative AI instructional integration
Topic 4 – Topic 7	0.8849	School curriculum/STEM education and children’s experiential learning
Topic 3 – Topic 4	0.8835	Generative AI instructional integration and school curriculum/STEM education
Topic 0 – Topic 7	0.8653	Educator readiness/ethics and children’s experiential learning
Topic 0 – Topic 1	0.8641	Educator readiness/ethics and adoption/self-efficacy factors
Topic 0 – Topic 3	0.8596	Educator readiness/ethics and generative AI instructional integration

The high centroid similarities indicate that several topics were semantically close. This did not undermine the topic model because the purpose of the analysis was to identify dominant thematic

concentrations within a connected literature, not to produce fully independent clusters. High-similarity topic pairs were manually inspected for possible redundancy. For example, Topics 3 and 5 were semantically close, but they represented different emphases: Topic 3 captured broader generative AI literacy and instructional tool integration, while Topic 5 centered more specifically on ChatGPT, writing, language learning, and prompting. Similarly, Topics 0, 2, and 4 shared educational and ethical vocabulary but differed in emphasis across educator readiness, higher education integration, and school curriculum/STEM education.

Figure C1. Topic Similarity Heatmap

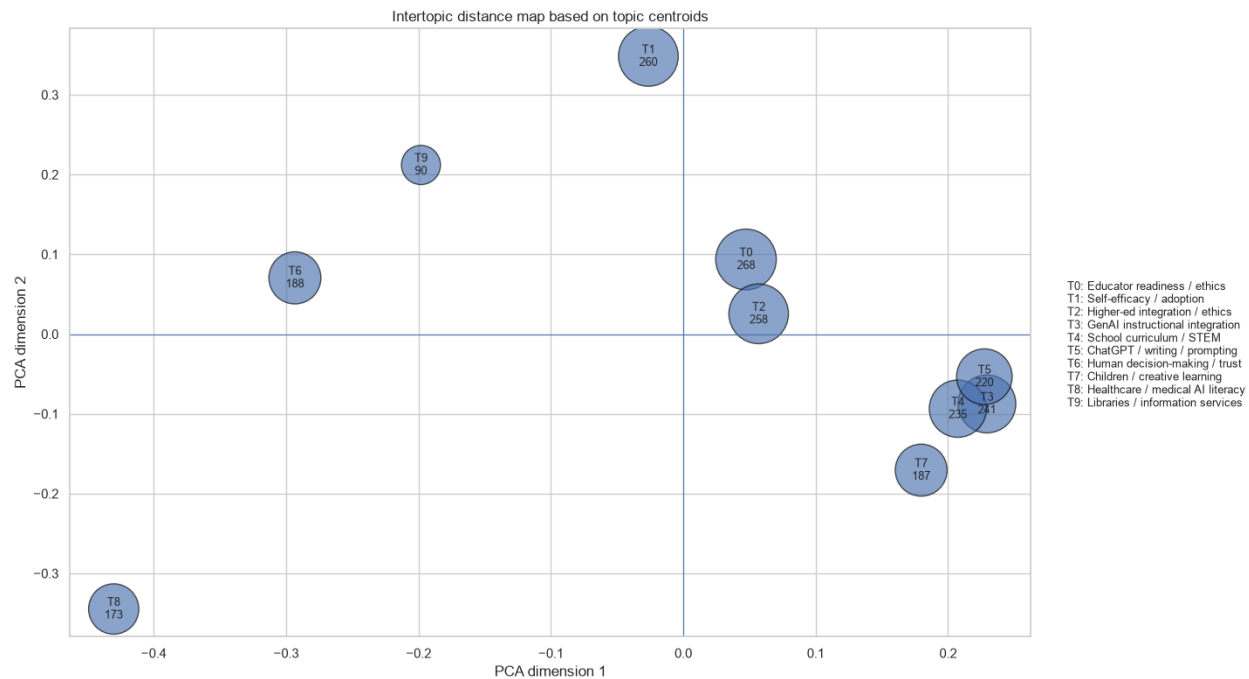


The heatmap was used to identify topic pairs requiring closer inspection. It supported the decision to retain the final topic structure while interpreting the topics as related thematic concentrations rather than strictly separated categories.

C.5 Intertopic Distance Map

An intertopic distance map was generated using topic centroids projected into two principal components. The map provides a visual representation of the relative proximity of topics in embedding space.

Figure C2. Intertopic Distance Map Based on Topic Centroids



The map shows that education- and generative-AI-related topics occupy a relatively compact area of the topic space, while healthcare, libraries/information services, human decision-making/trust, and self-efficacy/adoption factors appear more spatially separated. This visualization was used as a supplementary check on the semantic relationships observed in the centroid-similarity diagnostics.

C.6 Top-Word Overlap

Top-word overlap was examined to identify possible redundancy among topic representations. Jaccard overlap scores were calculated for top terms associated with each topic pair. The highest overlaps are shown in Table C2.

Table C2. Highest Top-Word Overlaps Across Topics

Topic pair	Jaccard overlap	Shared words
Topic 3 – Topic 5	0.333	critical; genai; generative; tools; writing
Topic 0 – Topic 4	0.250	curriculum; school; understanding
Topic 2 – Topic 3	0.176	challenges; critical; higher
Topic 2 – Topic 5	0.111	academic; critical

Topic pair	Jaccard overlap	Shared words
Topic 2 – Topic 6	0.111	digital; ethical
Topic 0 – Topic 2	0.111	educators; ethical
Topic 2 – Topic 9	0.111	academic; ethical
Topic 0 – Topic 7	0.111	children; understanding
Topic 5 – Topic 9	0.111	academic; tools
Topic 4 – Topic 7	0.111	design; understanding

The highest overlap occurred between Topic 3 and Topic 5, consistent with their shared focus on generative AI tools. The second-highest overlap occurred between Topic 0 and Topic 4, reflecting shared educational vocabulary around curriculum, school contexts, and understanding. These overlaps were judged acceptable because manual inspection indicated distinct thematic emphases within each pair.

C.7 Stability Across Random Seeds

To assess the stability of the KMeans-guided topic solution, clustering was repeated across multiple random seeds and pairwise similarity between solutions was calculated using adjusted Rand index and normalized mutual information. Across 45 pairwise comparisons, the mean adjusted Rand index was 0.6676 and the mean normalized mutual information was 0.7635.

Table C3. KMeans Stability Across Random Seeds

Statistic	Adjusted Rand Index	Normalized Mutual Information
Count	45	45
Mean	0.6676	0.7635
Standard deviation	0.0797	0.0448
Minimum	0.5584	0.7002
25th percentile	0.6092	0.7285
Median	0.6466	0.7528
75th percentile	0.7195	0.7868
Maximum	0.8770	0.8908

These values indicate moderate stability for an embedding-based clustering solution using short academic texts. Some topic-boundary variation occurred across random seeds, but the overall thematic structure was reasonably robust for the study’s field-mapping purpose.

C.8 Hierarchical Topic Structure

BERTopic’s hierarchical topic output was used as an additional diagnostic to examine higher-order relationships among the ten topics. The hierarchy indicated several broader thematic branches.

Topics 3 and 5 formed a close branch, reflecting the proximity of generative AI literacy and ChatGPT-mediated writing/prompting. Topic 1 then joined this branch, linking adoption-oriented research to the broader generative AI and tool-use cluster. Topics 0 and 4 also formed a close branch, reflecting shared emphasis on school curriculum, educator readiness, and foundational AI understanding. Topics 0, 2, 4, and 7 subsequently formed a broader education-oriented branch. Topic 9, focused on libraries and information services, merged at the highest level, suggesting that it was the most distinct of the ten topics.

The hierarchical output was used to support interpretation of macro-level relationships among topics. It was not used to relabel or merge topics in the final model.

C.9 Manual Validation and Supplementary Diagnostic Files

A manual validation sample of 200 records was exported to support inspection of topic assignments and representative documents. In addition, diagnostic files were generated for topic-size diagnostics, topic-centroid similarity, centroid similarity pairs, top-word overlap pairs, KMeans seed stability, comparison of alternative topic-number solutions, hierarchical topics, topic-label lookup, intertopic distance mapping, topic hierarchy visualization, and topic similarity heatmapping.

These supplementary materials were used to assess whether the selected solution provided a defensible field-level representation of self-identified AI literacy research. The diagnostic outputs were not treated as substitutes for interpretation; rather, they were used alongside manual inspection of topic labels, representative terms, representative documents, and temporal patterns.